

Attacks on Sparse LWE and Sparse LPN with new Sample-Time tradeoffs

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Abstract

This paper extends the Kikuchi method to give algorithms for decisional k -sparse Learning With Errors (LWE) and k -sparse Learning Parity with Noise (LPN) problems for higher moduli q . We create a Kikuchi graph for a sparse LWE/LPN instance and use it to give two attacks for these problems. The first attack decides by computing the spectral norm of the adjacency matrix of the Kikuchi graph, which is a generalization of the attack for $q = 2$ given by Wein et. al. (Journal of the ACM 2019). The second approach computes non-trivial closed walks of the graph, and then decides by computing a certain polynomial of edge labels in the walks. This is a generalization of the attack for $q = 2$ given by Gupta et. al. (SODA 2026). Both the attacks yield new tradeoffs between sample complexity and time complexity of sparse LWE/LPN.

1 Introduction

The Learning With Errors (LWE) problem, introduced by Regev [Reg05], and its close cousin, the Learning Parity with Noise (LPN) problem [Ale03], form central hardness assumptions in modern cryptography. In LWE, samples from the distribution $\mathcal{A}_{s,\chi}$ on $\mathbb{Z}_q^n \times \mathbb{Z}_q$ are generated by selecting $\mathbf{a} \in \mathbb{Z}_q^n$ uniformly at random, an error $e \in \mathbb{Z}_q$ from a distribution χ (typically a discrete Gaussian with width $r < q$), and computing $(\mathbf{a}, \langle \mathbf{a}, \mathbf{s} \rangle + e)$ modulo q ; the decision version requires distinguishing these from uniform samples on $\mathbb{Z}_q^n \times \mathbb{Z}_q$. LPN, most commonly studied over $\mathbb{Z}_2$ with Bernoulli noise or more generally over modulus q with small hamming weight error, shares similar structure. Most lattice-based cryptosystems rely on LWE's conjectured intractability against classical and quantum algorithms [Reg05; Pei09; Gen09; BV14; Pei16], while LPN's hardness rests on decades of cryptanalysis [Gol00; CM01; Fei02; MST03; AOW15a; AL16a; KMOW17a], offering competitive efficiency despite fewer constructions [Pie12; ADINZ17a].

Variants of these problems are considered to balance security with efficiency of cryptographic primitives. Different choices of modulus, noise distribution, or structural constraints (such as sparsity) can lead to primitives with faster key generation, smaller ciphertexts, or reduced computational overhead, while still relying on problems which are believed to have same hardness [LPR13; ABW10]. In this work, we focus on LWE and LPN with sparsity constraints. A sparse version can be considered where the coefficient vectors $\mathbf{a} \in \mathbb{Z}_q^n$ have only k non-zero entries for a parameter

$k < n$. The motivation for this sparsity is that the samples can be computed and stored with a roughly $O(n/k)$ factor improvement in efficiency. Hardness of sparse LPN over $\mathbb{Z}_2$ and variants have been studied for as long as forty years in average-case complexity and have intimate connections to constraint satisfaction problems [Hås84; Gol11; AOW15b; AL16b; KMOW17b]. Sparse variants of LWE/LPN over large moduli q (where q is at least polynomial in the security parameter) are useful for designing efficient advanced cryptographic primitives, including FHE but remain underexplored. We summarize the current known attacks for sparse variants below.

The task in analyzing the hardness of sparse LPN over $\mathbb{Z}_2$ (and in fact, sparse LWE in general) is to determine the computational complexity as a function of m , i.e. the number of samples $(\mathbf{a}, b)$. For $m \leq n$, the random and the planted noisy instances are statistically indistinguishable. Raghavendra, Rao and Schramm [RRS17] showed that sparse LPN over $\mathbb{Z}_2$ can be solved in $n^{O(l)}$ time when m is a quantity of the form $l \left(\frac{n}{l}\right)^{k/2} \log^{c(k)}(n)$ using a degree l “Sum of squares” method for $l \geq k/2$. An improved (and the best known) bound of the form $l \left(\frac{n}{l}\right)^{k/2} \log n$ for m with the same time complexity can be obtained using the “Kikuchi method” by Wein, Alaoui, and Moore [WEAM19] (independently by Hastings [Has20]).¹ Schmidhuber, O’Donnell, Kothari, and Babbush [SOKB25] gave a quantum algorithm that achieves a quartic speedup in time (i.e. $n^{1/4} \cdot \text{poly}(n)$) over the Kikuchi method for certain choices of the sample complexity m , while using exponentially less space. More recently, Gupta, He, O’Donnell, and Singer [GHOS26] used the Kikuchi graph defined by [WEAM19] and gave a new classical algorithm with a nearly quadratic speedup over the Kikuchi Method for the same sample complexity.

Despite the presence of numerous other variants of LWE, sparse LWE was not studied until a recent work by Jain, Lin, and Saha [JLS24]. Assuming hardness of sparse LWE, the authors construct secret key Linearly Homomorphic Encryption (LHE) schemes with slightly super-constant/constant overheads, and secret key homomorphic encryption for arbitrary constant-degree polynomials with slightly super-constant/constant overheads. They also perform some cryptanalysis on sparse LWE by suggesting two attacks for sparse LWE, namely the sparse vector in kernel attack and the dense minor attack. On analyzing the feasibility of these attacks for $k \geq \Omega(\log n)$ and $m = \text{poly}(n)$, it turns out that these attacks will take exponential time in n with high probability. Hence, the authors conjecture that sparse LWE with sparsity $k \geq \Omega(\log n)$ and polynomially many samples $m = \text{poly}(n)$ is as hard as the standard LWE with dimension $n' = \Theta(n)$ and polynomially many samples.

Our Contributions. This work is towards a more fine-grained analysis of the complexity of sparse LWE and sparse LPN over higher moduli. In particular, we obtain tradeoffs between the sample complexity m and time complexity for sparse LWE and sparse LPN over higher moduli q .

Theorem 1.1. (Informal) For dimension n , number of samples $m \geq \left(\frac{n}{l}\right)^k$, modulus q , width parameter $r < q$, and a tradeoff parameter $k/2 \leq l \leq n$:

- (Theorem 5.20) There exists an algorithm that solves the decisional k -sparse LWE in $\tilde{O}\left(\left(\frac{n}{l}\right)q^l\right)$ time.
- (Theorem 6.15) If q is prime, $q \gg r(l \log n)^{3/2}$, there exists an algorithm that solves the decisional k -sparse LWE in $\tilde{O}\left(\sqrt{\left(\frac{n}{l}\right)q^l} (n/l)^k\right)$ time.

Theorem 1.2. (Informal) For dimension n , number of samples $m \geq \left(\frac{n}{l}\right)^k$, modulus q , probability mass for non-zero error $\mu = \frac{q-1}{q} - \frac{1}{\text{poly}(n)}$, and a tradeoff parameter $k/2 \leq l \leq n$:

¹The Kikuchi method is inspired from Kikuchi free energy, a marginal approximation to Gibbs free energy [Kik51]

- (Theorem 5.21) There exists an algorithm that solves the decisional k -sparse LPN in $\tilde{O}\left(\binom{n}{l}q^l\right)$ time.
- (Theorem 6.20) If q is prime, there exists an algorithm that solves the decisional k -sparse LPN in $\tilde{O}\left(\left(\binom{n}{l}q^l\right)^{1/2+\varepsilon} (n/l)^k\right)$ time for an arbitrarily small constant ε .

As can be seen, the parameter l in the above theorems trades sample complexity for time. As an interesting case, if either $k = O(\log n / \log \log n)$ or we are allowed $m = n^{\Omega(\log \log n)}$ samples, we can choose l appropriately so that the time complexity is $\exp(\delta n)$ for arbitrarily small $\delta > 0$. Thus, assuming standard LWE does not have an arbitrary small exponential time, our result also establishes the tightness of the above conjecture on the hardness of sparse LWE by [JLS24].

Our methodology is to define a k -LIN $_q$ instance as a generalization of a k -XOR instance, and extend the Kikuchi method used by [WEAM19] and more recently [GHOS26] to distinguish random vs planted instances of k -LIN $_q$. For both extensions, this involves formulating a large graph corresponding to the k -LIN $_q$ instance, called the Kikuchi Graph. The method corresponding to [WEAM19] involves analysing the spectral norm of adjacency matrix of this graph. The method corresponding to [GHOS26] involves finding a short, closed, ‘non-trivial’ walk in the graph and then analysing the absolute value of a polynomial formed by the edge labels in the closed walk. Table 1 describes the pros and cons of the two methods at a high level.

Spectral Method	Closed Walk Method
Time: $\tilde{O}\left(\binom{n}{l}q^l\right)$, Samples: $\geq \left(\frac{O(n)}{l}\right)^k$	Nearly quadratic speedup with same sample complexity when $l \gg k$
Works when left-hand sides of k -LIN $_q$ are uniform	Also works for arbitrary left-hand sides of k -LIN $_q$
Does not assume primality or lower bound on q	Assumes primality, for sparse LWE also lower bound on q
Works for general noise distribution	Works for bounded-error and LPN noise distribution
Has a quantum speedup	Quantum speedup not known

Table 1: Comparing Spectral and Closed Walk Methods

Related Work. We recently realized that our construction of Kikuchi graph for k -LIN $_q$, and the analysis for upper bound of spectral norm for the random instance is similar to that discussed by Peter Manohar in their PhD thesis (Chapter 13 of [Man24]).² However, both our construction and our analysis is more general, which creates a significant improvement in our results. Let us elaborate on this:

- Their analysis is only for deciding whether a k -LIN $_q$ instance is random or not. We solve the harder problem of deciding whether the instance is random or planted. For this, we do an additional analysis for *lower bound* of spectral norm for a planted instance.
- Their construction of the Kikuchi Matrix is for prime q , whereas ours doesn’t require primality of q .

²We did not find a published work of this though.

- The eligible edges in the Kikuchi graph are much more in our construction (see Construction 4.9) than theirs (see Definition 13.0.2, [Man24]). As a consequence, we attain a sample complexity requirement of $m \geq \left(\frac{O(n)}{l}\right)^k$ (see Corollary 5.17) which is much better for large q than theirs: $m \geq \left(\frac{O(qn)}{l}\right)^{k/2}$ (Theorem 13.0.1, [Man24]). For small q , we match their sample complexity (see Corollary 5.16).

Organization. Section 2 gives a high level overview of our attacks. Section 3 introduces some basic notations and the definitions of sparse LWE and sparse LPN over higher modulus q . Section 4 describes the Kikuchi graph construction for k -LIN $_q$. In Sections 5, 6, we utilize the Kikuchi graph to distinguish planted vs random k -LIN $_q$ instances using the spectral method and q -ary cover method respectively. Finally, in Section 7, we compare our attacks, as well as discuss the attack strategies by [JLS24].

2 Technical Overview

In this section we provide a high-level overview of Sections 4, 5, 6 which are our technical contributions. In particular, we generalize the works of [WEAM19] and [GHOS26] to higher modulus q .

2.1 Kikuchi Graphs for sparse LWE/LPN.

Given an instance of the k -XOR problem (k -sparse LWE/LPN for $q = 2$), [WEAM19] (and independently [Has20]) used the so-called Kikuchi Method to create an undirected graph called the Kikuchi graph. Table 2 summarizes how prior work converts a k -sparse instance to a Kikuchi graph for $q = 2$, alongside our generalization of this reduction for $q > 2$.

Step	Approach for $q = 2$	Approach for $q > 2$
Convert equations	from linear equations over $\mathbb{Z}_2$ to product equations over $\{\pm 1\}$	from linear equations over $\mathbb{Z}_q$ to product equations over $\Omega_q := \{e^{2\pi i s/q} : s \in \mathbb{Z}_q\}$
k -sparse to 2-sparse	lift to $\binom{[n]}{l}$ dimensions using symmetric difference of vertices	lift to $\binom{[n]}{l} \times [q]^l$ dimensions using <i>generalized</i> symmetric difference
Kikuchi Graph	undirected	directed graph ³ with edges: $T \xrightarrow{b} T', T' \xrightarrow{b^*} T$

Table 2: Differences in the Kikuchi Graph for $q = 2$ vs $q > 2$

Let us review the steps for $q = 2$ in more detail:

1. Linear equations over $\mathbb{Z}_2$ are rewritten as product equations over $\{\pm 1\}$.
2. Each k -sparse equation is converted into a set of 2-sparse equations, i.e., each equation is an equation in two variables. This is done by assigning variables to l -sized subsets of $[n]$ where

³For our notion of generalized symmetric difference the Kikuchi graph is directed, which will be also be required in the analysis of spectral method and q -ary cover.

$l \geq k/2$ is a parameter. For each original equation of the form $C : \prod_{j \in S} x_j = b$, we write down equations $C' : y_T \cdot y_{T'} = b$ for all possible l -sized subsets $T, T' \subseteq [n]$ such that their symmetric difference equals S . For every vector $\mathbf{s} \in \{\pm 1\}^n$, one can define a natural vector $\mathbf{s}^{\odot l} \in \{\pm 1\}^{\binom{[n]}{l}}$ such that $\mathbf{s}$ satisfies C iff $\mathbf{s}^{\odot l}$ satisfies C' .

3. We note that a 2-sparse set of equations corresponds to a graph with variables as nodes and an edge between each pair of variables that appear in an equation. The weight of the edge is the value b on the RHS of the equation.

Now we discuss the specific modifications we had to make in order to generalize this for higher q :

- i. Linear equations over $\mathbb{Z}_q$ are rewritten as product equations over $\Omega_q = \{1, \omega_q, \dots, \omega_q^{q-1}\}$, where $\omega_q = e^{2\pi i/q}$ is a primitive q^{th} root of unity.
- ii. To successfully convert a k -sparse equation to a collection of 2-sparse equations, there are a couple of challenges in the $q > 2$ regime that we mitigate. Firstly, to describe the LHS of an equation, the set of non-zero variables is not sufficient; we also need to encode the coefficients of the non-zero variables. Accordingly, the high-dimensional space will also have to be l -sized subsets of index-coefficient pairs, i.e., $\binom{[n]}{l} \times [q]^l$. Secondly, for defining the 2-sparse constraints in this space, we will generalize the notion of choosing pairs satisfying a symmetric difference. Construction 4.9 describes this transformation.
- iii. Our generalization of symmetric difference is asymmetric and yields a directed graph. To generalize the spectral method, it is desirable that the adjacency matrix of the graph be *Hermitian*. To generalize the even cover method, it is desirable that the in-degree and out-degree of every vertex is the same. We mitigate both these issues by adding edges $T' \xrightarrow{b^*} T$ for every original edge $T \xrightarrow{b} T'$ in the graph.

2.2 The spectral method

After constructing the Kikuchi graph, the spectral method for both $q = 2$ and ours for $q > 2$ proceeds as follows:

The spectral norm of the Kikuchi graph (its adjacency matrix) has expectation 0 for a random instance. In the planted case, the expectation is $\geq \rho\Delta$, where Δ is the average degree of the graph and ρ is related to the variance of the noise distribution. In order to distinguish between a random instance and a planted instance, appropriate tail bounds are proved to show that the output of the algorithm will be unambiguously separated in the two cases.

The method we employed for the tail bounds slightly deviates from the ones used for $q = 2$. In the latter, the upper tail bound for the random case was obtained using Matrix Chernoff bounds for Rademacher series, and the lower tail bound for the planted case was obtained using properties of Skellam distribution. We use the *Matrix Bernstein inequality* to get both the tail bounds.⁴

2.3 The q -ary cover method

[GHOS26] solved the k -XOR problem via a usage of the Kikuchi Graph that is different from the way it is used in the spectral approach.

⁴it may be possible to derive generalizations of both techniques used for k -XOR, which could give slightly better tail bounds, but here we use the Matrix Bernstein inequality since it is readily available and serves the purpose

Approach for $q = 2$	Approach for prime $q > 2$
Consider an even cover C as a subset of constraints.	Define a q -ary cover $\mathbf{c}$ as a multiset containing constraints / vector in $\mathbb{Z}_q^m$.
Each variable in even cover occurs even times.	Coefficients of each variable sum to 0 mod q (with multiplicities).
Distinguish planted vs random by proving an anti-concentration result for a multi-linear polynomial of random variables.	Solve LPN problem using an analogous anti-concentration result. Solve LWE problem via boundedness of $\mathbf{c}^T \mathbf{b}$ where $\mathbf{b} \in \mathbb{Z}_q^m$ is the right-hand side.

Table 3: Differences in the q -ary cover method for $q = 2$ vs prime $q > 2$

1. They consider an *even cover* of the k -XOR instance, which has been studied in several works (eg. [FKO06; Fei08; GKM22]). An even cover is a subset of constraints such that every variable occurs an even number of times in the subset. Equivalently, an even cover corresponds to a subset of linearly dependent left-hand sides in the k -XOR instance.
2. They relate an even cover of a k -XOR instance with a closed walk in its Kikuchi graph. For some $\varepsilon > 0$ they find $O(n^{\varepsilon l})$ even covers (with high probability) which are non-empty, distinct and short sized ($O(l \log n)$).
3. For an even cover C , they consider the product of right-hand sides of constraints ψ_C . This has expectation 0 in the random case, and $\rho^{|C|}$ in the planted case.
4. To solve the distinguishing problem, they consider $|\sum_{C \in \mathcal{C}} \psi_C|$ for the collection $\mathcal{C}$ of non-empty, distinct, short covers. One needs to show a separation for this quantity in the two cases with high probability. The summands ψ_C are not independent, as distinct even covers may still have overlapping constraints. This prevents us from using standard concentration inequalities. For the random case, they are able to use the second moment method to show an upper bound on $|\sum_{C \in \mathcal{C}} \psi_C|$. For the planted case, they show *anti-concentration* of $|\sum_{C \in \mathcal{C}} \psi_C|$ around 0 by developing anti-concentration results for a certain class of multi-linear polynomials.

Let us discuss how we extend this method for prime $q > 2$:

- i. We define a q -ary cover of a k -sparse instance, as a *multiset* of constraints from the instance. In this case, for every index, the sum of its coefficients (counted with multiplicities in the multiset) must equal 0 mod q . We emphasize that we need to allow a q -ary cover to be a multiset to find (or even claim existence of) one given the sample complexities we are interested in. Again, a q -ary cover is equivalent to a subset of linearly dependent left-hand sides ($A \in \mathbb{Z}_q^{m \times n}$) in the LWE/LPN instance. It corresponds to $\mathbf{c} \in \mathbb{Z}_q^m$ such that $\mathbf{c}^T A = 0$, and c_i corresponds to the multiplicity of the i^{th} constraint from A .
- ii. Like [GHOS26], we too are able to find many non-trivial, distinct, short q -ary covers via finding closed walks in the Kikuchi Graph that we define.
- iii. For the LPN problem over prime $q > 2$, we too consider the product of right-hand sides of the constraints in a q -ary cover $\mathbf{c} \in \mathbb{Z}_q^m$. This time the product of right-hand sides of constraints $\psi_{\mathbf{c}}$ has terms $\eta_i^{c_i}$, for noise η_i in each independent i^{th} constraint.

Fortunately, the noise distribution η in LPN problem is uniformly distributed over $\Omega_q \setminus \{1\}$, so $\eta^s \equiv \eta$ for any $s \in \mathbb{Z}_q \setminus \{0\}$. Thus, $\psi_{\mathbf{c}}$ is identical to a multilinear monomial of independent noise variables η_i 's, and we are able to use the same separation techniques as in $q = 2$.

- iv. For the LWE problem, with a discrete Gaussian distribution $\psi_{\mathbf{c}}$ is not multilinear, and we are not able to get a good lower bound for $\mathbb{E}[\psi_{\mathbf{c}}]$ in the planted case. So we are not able to apply this approach. However, we are actually able to give a different attack which is *simpler*, thanks to the boundedness of the noise with high probability.

Table 3 compares the key differences between the even cover method by [GHOS26] and our q -ary cover method.

3 Preliminaries

Notation. Let $\mathbb{N}, \mathbb{R}, \mathbb{C}$ denote the set of naturals, reals, complex numbers respectively. For $n \in \mathbb{N}$, $[n] := \{1, 2, \dots, n\}$. For $0 \leq r \leq n$, $\binom{[n]}{r}$ denotes the set of all r -size subsets of $[n]$. For $a \in \mathbb{C}$, a^* denotes its complex conjugate. For a vector $\mathbf{x}$, unless stated otherwise, x_i denotes the i^{th} coordinate of $\mathbf{x}$. For a matrix A , $\|A\|$ denotes its spectral norm.

3.0.1 Sparse LWE and Sparse LPN.

For any $r > 0$, define $\rho_r(x) = \exp(-\pi x^2/r^2)$ for all $x \in \mathbb{R}$. We write ρ for ρ_1 . For a discrete set $S \subseteq \mathbb{R}$, we extend ρ to sets by $\rho_r(S) = \sum_{x \in S} \rho_r(x)$.

Let us recall the discrete Gaussian distribution on the integers and $\mathbb{Z}_q$.

Definition 3.1. *The discrete Gaussian distribution on the integers with width r , denoted $D_{\mathbb{Z},r}$, is the distribution having probability mass at $x \in \mathbb{Z}$ proportional to $\rho_r(x)$.*

Definition 3.2. *The discrete Gaussian distribution on $\mathbb{Z}_q$ with width r , denoted $D_{\mathbb{Z}_q,r}$ is the distribution having probability mass at $j \in \mathbb{Z}_q$ proportional to*

$$\sum_{x \in q\mathbb{Z}+j} \rho_r(x) = \sum_{x \in q\mathbb{Z}+j} e^{-\pi x^2/r^2}$$

We now recall the k -sparse LWE problem from [JLS24].

Definition 3.3. (Definition 4.1 in [JLS24]) *For a dimension $n \in \mathbb{N}$, a sparsity parameter $k = k(n)$, a modulus q , we define the distribution $D_{\text{coeff},n,k,q}$ that first samples a uniformly random set $S \in \binom{[n]}{k}$. Then, the distribution samples a random vector as follows. Sample $\mathbf{a} \in \mathbb{Z}_q^n$ so that each coordinate a_i is set to 0 if $i \notin S$. Otherwise, a_i is chosen at random from $\mathbb{Z}_q$.*

Remark 3.4. In Definition 3.3, we are choosing k indices which may or may not get non-zero coefficients since their coefficients are chosen uniformly from $\mathbb{Z}_q$ instead of $\mathbb{Z}_q \setminus \{0\}$. So a sample from $D_{\text{coeff},n,k,q}$ has at most k -non-zero coefficients. This is slightly different from a k -XOR sample where there were exactly k non-zero coefficients. However, either definition won't make much difference in our results, and we stick to the definition given in [JLS24].

Definition 3.5. (Definition 4.3 in [JLS24]) *For an integer dimension n , sparsity parameter k , sample complexity m , modulus q , and noise parameter $r = r(n)$, we define the k -LWE $_{n,m,q,r}$ problem as a distinguishing problem that is given samples from either of two distributions, and is required to guess the distribution. For the first (planted) distribution, we generate m samples as follows:*

- Sample m coefficient vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$ where each $\mathbf{a}_i \leftarrow D_{\text{coeff}, n, k, q}$.
- Sample a secret vector $\mathbf{s} \leftarrow \mathbb{Z}_q^n$.
- Sample m noise values from $\{e_i \leftarrow D_{\mathbb{Z}_m, r}\}_{i \in [m]}$.
- The first distribution consists of $\{(\mathbf{a}_i, \beta_i = \langle \mathbf{s}, \mathbf{a}_i \rangle + e_i \bmod q)\}$.

The second (random) distribution is exactly the same except $\{\beta_i\}_{i \in [m]}$ are randomly sampled from $\mathbb{Z}_q$.

The standard decisional LWE problem is identical to a sparse LWE distinguishing problem as above except that the vectors $\mathbf{a}_i$ are chosen randomly from $\mathbb{Z}_q^n$ as opposed to being sampled as sparse vectors.

We also define a sparse version of the Learning Parity with Noise (LPN) problem for higher modulus, which (the non-sparse version) was defined in [ADINZ17b].

Definition 3.6. For an integer dimension n , sparsity parameter k , sample complexity m , modulus q , and noise parameter $\mu = \mu(n)$, we define the k -LPN $_{n, m, q, \mu}$ problem as a distinguishing problem that is given samples from either of two distributions, and is required to guess the distribution. For the first (planted) distribution, we generate m samples as follows:

- Sample m coefficient vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$ where each $\mathbf{a}_i \leftarrow D_{\text{coeff}, n, k, q}$.
- Sample a secret vector $\mathbf{s} \leftarrow \mathbb{Z}_q^n$.
- Sample m noise values from $\{e_i \leftarrow D_\mu\}_{i \in [m]}$, where $D_\mu(0) = 1 - \mu$, $\forall j \in \mathbb{Z}_q \setminus \{0\} D_\mu(j) = \mu/(q-1)$.
- The first distribution consists of $\{(\mathbf{a}_i, \beta_i = \langle \mathbf{s}, \mathbf{a}_i \rangle + e_i \bmod q)\}$.

The second (random) distribution is exactly the same except $\{\beta_i\}_{i \in [m]}$ are randomly sampled from $\mathbb{Z}_q$.

4 Kikuchi Graph for sparse LWE/LPN instances

4.1 Product equations over q^{th} roots of unity

We start by viewing a linear equation over $\mathbb{Z}_q$ as a product equation over $\mathbb{C}$. Looking ahead, such a transformation is not merely a notational preference; in Section 5 we want to deal with matrices over $\mathbb{C}$ where we can utilize the well-studied theory of eigenvalue estimation for Hermitian matrices, and in Section 6, to solve sparse LPN we want to analyze products of right hand sides which is more convenient when we wish to utilize the independence of constraints.

The subgroup of $(\mathbb{C}, \cdot)$ isomorphic to $\mathbb{Z}_q$ is $\Omega_q = \{1, \omega_q, \dots, \omega_q^{q-1}\}$, where $\omega_q = e^{2\pi i/q}$ is a primitive q^{th} root of unity. Thus, an equation in $\mathbb{Z}_q$ of the form

$$a_{i_1} t_{i_1} + \dots + a_{i_k} t_{i_k} = p_i$$

can be converted to

$$x_{i_1}^{a_{i_1}} \dots x_{i_k}^{a_{i_k}} = b_i$$

where $b_i \in \Omega_q$, $a_{i_j} \in [q]$ (equivalently in $\mathbb{Z}_q$) and variables x_{i_j} take values in Ω_q for all $j \in [k]$. The LHS can be represented by the set $S_i = \{(i_j, a_{i_j}) : j \in [k]\}$.

Definition 4.1. A k -LIN $_q$ instance $\mathcal{I}$ over variables indexed by $[n]$ consists of a multiset of constraints $\mathcal{C} = (S, b)$, where each scope $S \in \binom{[n]}{k} \times [q]^k$ and each right-hand side $b \in \Omega_q$.

In the decisional LWE and LPN problems, we are concerned with the average-case problem of distinguishing a system of uniformly random linear equations from a system of linear equations where a solution is planted with some noise. Let us formally define these two types of instances in the setting of k -LIN $_q$. These are analogous to the random and planted instance definitions in ([SOKB25], Notations 5,6).

Definition 4.2. We let $\mathcal{R}_{n,k}(m)$ denote the distribution of uniformly random k -LIN $_q$ instances in which m scopes $S_1, \dots, S_m \in \binom{[n]}{k} \times [q]^k$ are chosen uniformly at random, and the associated right-hand sides $b_1, \dots, b_m$ are i.i.d. uniform random variables over Ω_q .

In the planted case, the right-hand sides are instead correlated with a “secret” assignment.

Notation 4.3. For a set $S \in \binom{[n]}{r} \times [q]^r$ for some $r \in [n]$, $\mathbf{x} \in \Omega_q^n$, denote

$$\mathbf{x}^S := \prod_{(j,a) \in S} x_j^a$$

Definition 4.4. Let $\mathcal{D}_\rho$ be a distribution on Ω_q such that for $X \sim \mathcal{D}_\rho$, $\mathbb{E}[X] = \rho$ for some $\rho \in [0, 1]$. Fix $\mathbf{z} \in \Omega_q^n$. We let $\mathcal{P}_{n,k}^{\mathbf{z}}(m, \mathcal{D}_\rho)$ denote the distribution of planted noisy k -LIN $_q$ instances in which m scopes $S_1, \dots, S_m \in \binom{[n]}{k} \times [q]^k$ are chosen uniformly at random, and the associated right-hand sides are $\eta_1 \mathbf{z}^{S_1}, \dots, \eta_m \mathbf{z}^{S_m}$, where $\eta_1, \dots, \eta_m \stackrel{iid}{\sim} \mathcal{D}_\rho$.

Remark 4.5. In the above definition, we have by design that ρ is real and in $[0, 1]$. This will be satisfied for ‘nice enough’ distributions like the noise models in LWE and LPN. When $\rho = 1$, $\mathcal{D}_\rho$ must give all probability mass to $1 \in \Omega_q$, so we simply write $\mathcal{P}_{n,k}^{\mathbf{z}}(m)$. The distribution $\mathcal{R}_{n,k}(m)$ is the same as $\mathcal{P}_{n,k}^{\mathbf{z}}(m, \mathcal{U})$, where $\mathcal{U}$ is the uniform distribution on Ω_q ($\rho = 0$) and this distribution is independent of $\mathbf{z}$.

It is the planted noisy k -LIN $_q$ instance where the definitions of LWE and LPN differ. The noise distribution of a planted LWE instance is typically a centered discrete Gaussian over $\mathbb{Z}$ which can be converted to a distribution over $\mathbb{Z}_q$ by appropriately transferring the probability masses of each integer point to its residue modulo q . The noise distribution in a planted LPN instance has its mass on $\mathbb{Z}_q \setminus \{0\}$ uniformly distributed to a small total value, and the rest goes to $0 \in \mathbb{Z}_q$. Both these distributions can be specified by a single parameter (for the discrete Gaussian, it is typically the width/variance). We consider two types of distributions over Ω_q corresponding to the planted distributions of LPN and LWE problems as in Definitions 3.6, 3.5 respectively.

Definition 4.6. For $\rho \in [0, 1]$, let $\mathcal{LPN}_\rho$ be the distribution on Ω_q such that for $\eta \sim \mathcal{LPN}_\rho$, $\mathbb{E}[\eta] = \rho$ and $\eta = \omega_q^e$ where $e \sim D_\mu$ as given in Definition 3.6. This means that $\rho = \sum_{a \in \mathbb{Z}_q} \omega_q^a D_\mu(a) = 1 - \mu + \frac{\mu}{q-1} \sum_{a \in \mathbb{Z}_q \setminus \{0\}} \omega_q^a = 1 - \frac{q\mu}{q-1}$.

Definition 4.7. For $\rho \in [0, 1]$, let $\mathcal{LWE}_\rho$ be the distribution on Ω_q such that for $\eta \sim \mathcal{LWE}_\rho$, $\mathbb{E}[\eta] = \rho$ and $\eta = \omega_q^e$ where $e \sim D_{\mathbb{Z}_q, r}$. It can be shown (see Appendix A Remark A.2) that $\rho \geq e^{-\pi r^2/q^2}$. In particular, for $r < q$, ρ is a constant.

4.2 The Kikuchi graph for k -LIN $_q$

We will now develop the following insight: viewing a k -sparse equation (a k -uniform hyperedge) as edges of a graph on a large vertex space. Given a k -LIN $_q$ instance $\mathcal{I}$, we wish to produce a directed graph, namely $\mathcal{K}_l(\mathcal{I})$ (for some parameter l). This technique was introduced for the k -XOR problem by Wein. et. al. [WEAM19] (and implicitly by Hastings [Has20]). Having a graph representative of the instance allows us to use either combinatorial or spectral properties of the graph for the distinguishing problem.

For a head start, here is some intuition on how this is done: Let us encode each possible $k/2$ -sized subset $T = \{(j, a) : j \in [n], a \in [q]\} \in \binom{[n]}{k/2} \times [q]^{k/2}$ as an index in the transformed space (assume for simplicity for now that k is even). Then an equation of the form $C : x_{i_1}^{a_{i_1}} \dots x_{i_k}^{a_{i_k}} = b_i$ can be converted to $C' : y_I^{-1} y_J = b_i$ where $I = \{(i_1, -a_{i_1}), \dots, (i_{k/2}, -a_{i_{k/2}})\}$, $J = \{(i_{k/2+1}, a_{i_{k/2+1}}), \dots, (i_k, a_{i_k})\}$. For $\mathbf{s} \in \Omega_q^n$, let $\tilde{\mathbf{s}} \in \Omega_q^{\binom{[n]}{k/2} \times [q]^{k/2}}$ be defined such that the T^{th} coordinate is $\mathbf{s}^T$. Then $\mathbf{s}$ satisfies equation C iff $\tilde{\mathbf{s}}$ satisfies C' . C' in turn can be viewed as an edge $I \xrightarrow{b_i} J$ with a sign b_i .

The above choice of I, J is only one of $\binom{k}{k/2}$ possible choices; we can choose any of the $k/2$ indices for I . Further, we need not restrict ourselves to $(k/2)$ -sized subsets, we can have l -sized subsets for an appropriate $l \geq k/2$. In this case, we not only have freedom to choose indices, but also the exponents corresponding to each index.

Notation 4.8. For an index $j \in [n]$, $S \in \binom{[n]}{r} \times [q]^r$ for some $r \in [n]$, if $(j, a) \in S$ for some (unique) $a \in [q]$, then define $\text{coeff}_S(j) := a$, else define $\text{coeff}_S(j) := 0$.

Construction 4.9. Let $\mathcal{I}$ be a k -LIN $_q$ instance, and $k/2 \leq l \leq n$ be a parameter. $\mathcal{K}_l(\mathcal{I})$ is a directed graph over vertices indexed by $[N] := \binom{[n]}{l} \times [q]^l$. For every constraint $(S, b) \in \mathcal{I}$ ⁵, $\mathcal{K}_l(\mathcal{I})$ has edges $T_1 \xrightarrow{b} T_2$ and $T_2 \xrightarrow{b^*} T_1$ for all $T_1, T_2 \in [N]$ which satisfy the following:

$$\forall j \in [n], \text{coeff}_{T_2}(j) - \text{coeff}_{T_1}(j) = \text{coeff}_S(j) \quad (1)$$

Remark 4.10. It can be seen that given $T_1, T_2 \in \binom{[n]}{l} \times [q]^l$, there is a unique S (not necessarily in $\binom{[n]}{k} \times [q]^k$) which satisfies (1).

Notation 4.11. Let us use the notation $\mathcal{S}(T_1, T_2) := S$ to denote this unique set. For $b \in \Omega_q$, let us also extend this notation to say $\mathcal{S}(T_1, T_2, b) := (S, b)$.

Remark 4.12. Given $S \in \binom{[n]}{k} \times [q]^k$ and $T_1 \in \binom{[n]}{l} \times [q]^l$, there exists at most one $T_2 \in \binom{[n]}{l} \times [q]^l$ such that $\mathcal{S}(T_1, T_2) = S$. Similarly, given $S \in \binom{[n]}{k} \times [q]^k$ and $T_2 \in \binom{[n]}{l} \times [q]^l$, there exists at most one $T_1 \in \binom{[n]}{l} \times [q]^l$ such that $\mathcal{S}(T_1, T_2) = S$.

For a constraint $(S, b) \in \mathcal{I}$, how many corresponding constraints does $\mathcal{K}_l(\mathcal{I})$ have? Before venturing to compute this, let us rewrite Construction 4.9 in a more suitable way.

Notation 4.13. For $S \in \binom{[n]}{r} \times [q]^r$ for some $r \in [n]$, we write $(i, *) \notin S$ to say that for all $a \in \mathbb{Z}_q$, $(i, a) \notin S$. Similarly, we write $(i, *) \in S$ to say that for some $a \in \mathbb{Z}_q$, $(i, a) \in S$.

Construction 4.14. (Equivalent to Construction 4.9) For every constraint $(S, b) \in \mathcal{I}$, $\mathcal{K}_l(\mathcal{I})$ has all edges (T_1, T_2, b) , (T_2, T_1, b^*) which satisfy the following:

⁵for simplicity, we assume that $\mathcal{I}$ does not have $(S, b), (S, b')$ for the same LHS S ; otherwise, we can distinguish the planted vs random instances by a simple analysis of the RHS as mentioned in the Introduction.

1. For every $(i, *) \notin S$, we have $(i, a) \in T_1 \cap T_2$ for at most one $a \in \mathbb{Z}_q$.

2. For every $(i, a) \in S$, one of the following holds:

(a) $(i, -a \bmod q) \in T_1, (i, *) \notin T_2$

(b) $(i, *) \notin T_1, (i, a) \in T_2$

(c) For $a', a'' \in \mathbb{Z}_q$ such that $-a' + a'' \equiv a \pmod{q}$, $(i, a') \in T_1, (i, a'') \in T_2$

Let $\theta^{(n,l,k)}$ be the number of edges added in $\mathcal{K}_l(\mathcal{I})$ corresponding to a constraint $(S, b) \in \mathcal{I}$. It can be computed in the following way: We'll start by choosing tuples to satisfy 2(c) of Construction 4.14, say r of them, and also each of the q possible choices of the exponents a', a'' . Then we need to choose the remaining $(k - r)$ tuples for condition 2(a), 2(b); for each condition, we will need to choose an equal number of disjoint set of tuples⁶, which amounts to choosing $(k - r)/2$ tuples. For case 1, there are $l - (r + (k - r)/2)$ elements to be chosen out of $n - k$ elements, each of which can have the exponent $a \in \mathbb{Z}_q$. Finally for each choice of (T_1, T_2) there are two directed edges. Thus we get the number of constraints for each original constraint as:⁷

$$\theta^{(n,l,k)} = 2 \sum_{r=0}^k \binom{k}{r} q^r \binom{k-r}{(k-r)/2} \binom{n-k}{l-(k+r)/2} q^{l-(k+r)/2} \quad (2)$$

We can obtain two lower bounds for $\theta^{(n,l,k)}$ by either having no tuples of 2(c) (i.e. $r = 0$), or all of them (i.e. $r = k$)⁸:

$$\text{Assuming } k/2 \leq l \leq n - k/2, \quad \theta^{(n,l,k)} \geq 2 \binom{k}{k/2} \binom{n-k}{l-k/2} q^{l-k/2} \quad (3)$$

$$\text{Assuming } k \leq l \leq n, \quad \theta^{(n,l,k)} \geq 2 \binom{n-k}{l-k} q^l \quad (4)$$

Depending on whether q is small or large, we will later utilize either of these bounds for the sample complexity. It can be seen by a routine calculation that upto a factor of $\exp(O(k))$, these lower bounds are tight.

5 Distinguishing by spectral method over Kikuchi Graphs

The Kikuchi Graph $\mathcal{K}_l(\mathcal{I})$ can be naturally associated with its adjacency matrix (for simplicity, we use the same notation to denote the adjacency matrix). By Construction 4.9, for every edge $T_1 \xrightarrow{b} T_2$, we have the edge $T_2 \xrightarrow{b^*} T_1$ present in the graph. Therefore $\mathcal{K}_l(\mathcal{I})$ is Hermitian. We will now proceed to determine whether $\mathcal{I}$ is planted or random by computing the spectral norm of $\mathcal{K}_l(\mathcal{I})$.

⁶this is because items 1 and 2(c) choose equal number of elements for both T_1, T_2 and we must maintain $|T_1| = |T_2| = l$

⁷in this equation we use the convention that if s is not a non-negative integer or $s > t$, then $\binom{t}{s} = 0$

⁸assume k is even, otherwise we can get similar bounds using $r = 1, k - 1$

5.1 Independent Summands of $\mathcal{K}_l(\mathcal{I})$

Firstly for a $k\text{-LIN}_q$ instance $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ we write $\mathcal{K}_l(\mathcal{I})$ as a sum of independent Hermitian matrices:

$$\mathcal{K}_l(\mathcal{I}) = \sum_{i \in [m]} \mathbf{X}_i \quad (5)$$

$$\mathbf{X}_i(T_1, T_2) := \begin{cases} b_i & \text{if } \mathcal{S}(T_1, T_2) = S_i \\ b_i^* & \text{if } \mathcal{S}(T_2, T_1) = S_i \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

By definition, $\mathbf{X}_i$ are Hermitian, and independent as the constraints in $\mathcal{I}$ are independent.

We now prepare to establish some key properties of $\mathbf{X}_i$. Observe that the graph corresponding to $\mathbf{X}_i$ is a collection of disjoint *chains* (see illustration in Figure 1). This is because from Remark 4.12, the graph corresponding to $\mathbf{X}_i$ never contains constraints (T_1, T_2, b_i) and (T'_1, T'_2, b_i) with $(T_1 = T'_1 \wedge T_2 \neq T'_2)$, or $(T_1 \neq T'_1 \wedge T_2 = T'_2)$.

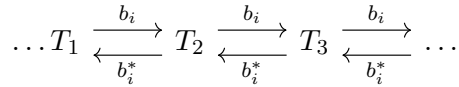


Figure 1: A chain in the graph of $\mathbf{X}_i$

Let us prove an upper bound on the spectral norm of the adjacency matrix of a signed graph. This is analogous to the well known spectral norm bound of an unsigned graph.

Lemma 5.1. *Let $\mathcal{G}$ be a graph where each edge is of the form (I, J, b) with $b \in \Omega_q$. Then $\|\mathcal{G}\| \leq D(\mathcal{G})$.*

Proof. Let ϕ be an eigenvector corresponding to the eigenvalue of $\mathcal{G}$ whose absolute value is $\|\mathcal{G}\|$. Let I be a vertex in $\mathcal{G}$ such that $|\phi(I)| \geq |\phi(J)|$ for all vertices J of $\mathcal{G}$. Since $\phi \neq \mathbf{0}$, we have $|\phi(I)| \neq 0$. Now

$$\begin{aligned} \|\mathcal{G}\| &= \frac{|(\mathcal{G}\phi)(I)|}{|\phi(I)|} = \frac{\left| \sum_{(I,J) \in E(\mathcal{G})} b \cdot \phi(J) \right|}{|\phi(I)|} \\ &\leq \frac{\sum_{(I,J) \in E(\mathcal{G})} |\phi(J)|}{|\phi(I)|} \leq \text{out-deg}(I) \leq D(\mathcal{G}) \end{aligned}$$

□

We state an upper bound for $\|\mathbf{X}_i\|$.

Lemma 5.2. *Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be any $k\text{-LIN}_q$ instance, and $\mathbf{X}_i$ be as defined in (6). Then $\|\mathbf{X}_i\| \leq 2$.*

Proof. From Remark 4.12, (the graph corresponding to) $\mathbf{X}_i$ never contains constraints (T_1, T_2, b_i) and (T'_1, T'_2, b_i) with $(T_1 = T'_1 \wedge T_2 \neq T'_2)$, or $(T_1 \neq T'_1 \wedge T_2 = T'_2)$. Thus, for a vertex T_1 , there are only 4 possible edges incident to T_1 :

$$(T_1, T_2, b_i), (T_2, T_1, b_i^*), (T'_1, T_1, b_i), (T_1, T'_2, b_i^*)$$

In particular, T_1 has an out-degree of at most 2. Thus the result follows from Lemma 5.1. □

Next, we state a result regarding sum of squares of $\mathbf{X}_i$. Looking ahead, this will be useful to compute the variance statistic corresponding to $\mathcal{K}_l(\mathcal{I})$. The key observation is that $\mathbf{X}_i^2$ can be viewed as an adjacency matrix of a graph that captures “2-hops” of the original graph of $\mathbf{X}_i$.

Notation 5.1. For a directed graph $\mathcal{G}$, let $\text{out-deg}_{\mathcal{G}}(T)$ denote the number of outgoing edges of a vertex T of $\mathcal{G}$, and let $D(\mathcal{G})$ denote the maximum outgoing degree of $\mathcal{G}$.

Lemma 5.3. Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be any k -LIN $_q$ instance, and $\mathbf{X}_i$ be as defined in (6). Then $\left\| \sum_{i \in [m]} \mathbf{X}_i^2 \right\| \leq 2D(\mathcal{K}_l(\mathcal{I}))$.

Proof. In the graph of $\mathbf{X}_i$, consider any edge pair

$$\dots T_1 \xrightleftharpoons[b_i^*]{b_i} T_2 \dots$$

which is part of a chain. By the fact that each vertex in the graph of $\mathbf{X}_i$ has both out-degree and in-degree at most 2, the edge (T_1, T_2, b_i) can contribute to at most 2 outgoing edges of T_1 in the graph corresponding to $\mathbf{X}_i^2$ (One is $T_1 \xrightarrow{b_i} T_2 \xrightarrow{b_i^*} T_1$ and the other is possibly $T_1 \xrightarrow{b_i} T_2 \xrightarrow{b_i} T_3$).

Let $\mathcal{G}$ be the graph corresponding to $\sum_{i \in [m]} \mathbf{X}_i^2$. We then have $\text{out-deg}_{\mathcal{G}}(T_1) \leq 2 \cdot \text{out-deg}_{\mathcal{K}_l(\mathcal{I})}(T_1)$ for each vertex T_1 . The lemma now follows by applying Lemma 5.1 on $\mathcal{G}$. $\square$

Let us now obtain an upper bound on $D(\mathcal{K}_l(\mathcal{I}))$, for which we use the distribution of the ‘left-hand sides’, i.e. the fact that $S_i \stackrel{iid}{\leftarrow} \binom{[n]}{k} \times [q]^k$. Note that this is the setting in both the random and planted instances.

Lemma 5.4. Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be a k -LIN $_q$ instance where $S_i \stackrel{iid}{\leftarrow} \binom{[n]}{k} \times [q]^k$. Let $N = \binom{n}{l} q^l$. For any $T \in \binom{[n]}{l} \times [q]^l$, let $d_T^{(i)}$ be the random variable denoting the number of edges corresponding to (S_i, b_i) outgoing from T in $\mathcal{K}_l(\mathcal{I})$. Then $\mathbb{E} \left[d_T^{(i)} \right] = \frac{\theta(n, l, k)}{N}$.

Proof. Let us call a vertex pair $(T_1, T_2) \in [N]^2$ eligible if it can be obtained by applying Construction 4.9 to some (S, b) where $S \in \binom{[n]}{k} \times [q]^k$. By symmetry, there is an equal number of eligible tuples outgoing from any vertex in $\mathcal{K}_l(\mathcal{I})$. Let Φ be the number of eligible tuples, so there are Φ/N eligible tuples outgoing from T . Moreover, since S_i is chosen uniformly at random, each eligible tuple is equally likely to be added in $\mathcal{K}_l(\mathcal{I})$, say with probability p . Thus, the expected number of added tuples is given by $p\Phi$, which is equal to $\theta(n, l, k)$ since that is the (deterministic) number of tuples added for any choice of (S, b) . Finally, $\mathbb{E} \left[d_T^{(i)} \right] = \frac{p\Phi}{N} = \frac{\theta(n, l, k)}{N}$. $\square$

Proposition 5.5. Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be a k -LIN $_q$ instance where $S_i \stackrel{iid}{\leftarrow} \binom{[n]}{k} \times [q]^k$. Let $N = \binom{n}{l} q^l$, $\Delta = \frac{m\theta(n, l, k)}{N}$. Then for any $\delta > 0$, we have $D(\mathcal{K}_l(\mathcal{I})) \leq (1 + \delta)\Delta$ except with probability at most $N \cdot \exp\left(-\frac{\delta^2 \Delta}{2 + \delta}\right)$.

Proof. As a consequence of Lemma 5.4, $\text{out-deg}(T) = \sum_{i \in [m]} d_T^{(i)}$ satisfies

$$\mathbb{E}[\text{out-deg}(T)] = \frac{m\theta(n, l, k)}{N}$$

Further, $\text{out-deg}(T)$ is a sum of independent random variables, and $D(\mathcal{K}_l(\mathcal{I})) = \max_{T \in [N]} \text{out-deg}(T)$.

Thus we can apply a Chernoff bound followed by a union bound to prove the proposition. $\square$

5.2 An upper bound for the random instance

Let us first recall one result that bounds the norm of a sum of random Hermitian matrices, namely the Matrix Bernstein inequality (see, eg., [Tro12]).

Theorem 5.6. (Theorem 1.4 in [Tro12]) Consider a finite sequence $\{\mathbf{X}_i\}$ of independent, random, Hermitian matrices with dimension n . Assume that each random matrix satisfies $\mathbb{E}(\mathbf{X}_i) = \mathbf{O}$ and $\|\mathbf{X}_i\| \leq R$ almost surely. Then, for all $t \geq 0$,

$$\mathbb{P} \left[\left\| \sum_i \mathbf{X}_i \right\| \geq t \right] \leq n \cdot \exp \left(\frac{-t^2/2}{\sigma^2 + (Rt)/3} \right),$$

where

$$\sigma^2 := \left\| \sum_k \mathbb{E}(\mathbf{X}_k^2) \right\|.$$

We will show that when $\mathcal{I} \sim \mathcal{R}_{n,k}(m)$, an upper bound on $\|\mathcal{K}_l(\mathcal{I})\|$ holds with high probability. To establish this, let us apply the Matrix Bernstein inequality to our case. Recall that when $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]} \sim \mathcal{R}_{n,k}(m)$, S_i is a uniformly random k -sparse set of index-coefficient pairs, and b_i is uniformly chosen from Ω_q .

We start with analysing the summands $\mathbf{X}_i$'s further when $\mathcal{I}$ is a random instance.

Lemma 5.7. Let $\mathcal{I} \sim \mathcal{R}_{n,k}(m)$, and $\mathbf{X}_i$ be as defined in (6). Then $\mathbb{E}[\mathbf{X}_i] = \mathbf{O}$.

Proof. Since for every constraint $(S_i, b_i) \in \mathcal{I}$, b_i is chosen uniformly at random from Ω_q , any entry of $\mathbf{X}_i$ is either always 0 or takes a value in Ω_q with equal probability. The result follows as the values in Ω_q sum to 0. $\square$

With this, the antecedent of Theorem 5.6 is satisfied with $\|\mathbf{X}_i\| \leq 2$ (from Lemma 5.2) for all $i \in [m]$. Also, from Lemma 5.3, we have

$$\sigma^2 = \left\| \sum_{i \in [m]} \mathbb{E}(\mathbf{X}_i^2) \right\| \leq 2D(\mathcal{K}_l(\mathcal{I})) \quad (7)$$

Now we choose an appropriate value of t to limit the probability in Theorem 5.6 to be an inverse polynomial, say $N^{-\epsilon}$ for some choice of $\epsilon > 0$. Thus we must satisfy

$$N \cdot \exp \left(\frac{-t^2/2}{2D(\mathcal{K}_l(\mathcal{I})) + 2t/3} \right) \leq N^{-\epsilon} \quad (8)$$

By simplifying (8) we have the following upper bound for the maximum eigenvalue:

Corollary 5.8. Let $\mathcal{I} \sim \mathcal{R}_{n,k}(m)$, $N = \binom{n}{l} q^l$. Then for any $\epsilon > 0$, except with probability at most $N^{-\epsilon}$, we have

$$\|\mathcal{K}_l(\mathcal{I})\| \leq \frac{2\sqrt{(1+\epsilon)\ln N}}{3} \left(\sqrt{(1+\epsilon)\ln N} + \sqrt{9D(\mathcal{K}_l(\mathcal{I})) + (1+\epsilon)\ln N} \right) \quad (9)$$

When the left-hand sides $S_i \in \binom{[n]}{k} \times [q]^k$ are chosen uniformly at random (which is the case for $\mathcal{R}_{n,k}(m)$), Proposition 5.5 gives us that $D(\mathcal{K}_l(\mathcal{I}))$ is upper bounded by $(1 + \delta)\Delta$ with high probability, where $\Delta = \frac{m\theta^{(n,l,k)}}{N}$ is the average out-degree of $\mathcal{K}_l(\mathcal{I})$, and $\delta > 0$ is a constant. We can now combine Corollary 5.8 and Proposition 5.5 to get an upper bound on the spectral norm with respect to Δ . We would like to choose Δ (which in turn means choosing the number of constraints m in $\mathcal{I}$) so that we get $\|\mathcal{K}_l(\mathcal{I})\| \leq \delta\Delta$. Observe that in (9) we can impose

$$\Delta \geq \frac{16\delta + 12}{3\delta^2}(1 + \epsilon) \ln N$$

to get this desired bound for all $\delta > 0$. Also, the probability of failure in Proposition 5.5 is upper bounded by $N^{-\epsilon}$ for this choice.

Theorem 5.9. *Let $\mathcal{I} \sim \mathcal{R}_{n,k}(m)$ and $\Delta = \frac{m\theta^{(n,l,k)}}{N} \geq \frac{16\delta + 12}{3\delta^2}(1 + \epsilon) \ln N$ for some $\delta, \epsilon > 0$. Then we have $\|\mathcal{K}_l(\mathcal{I})\| \leq \delta\Delta$ except with probability at most $2N^{-\epsilon}$.*

Finally let us restate this result in terms of the parameters of $\mathcal{I}$. Here we will use the lower bounds of $\theta^{(n,l,k)}$ from (3), (4) depending on the value of q and l .

If $q = O(n/l)$, then (3) will give a better bound than (4). On the other hand, if $q = \omega(n/l)$, (4) gives a better bound.

Corollary 5.10. *Let k, n, l, q be positive integers such that $k/2 \leq l \leq n - k/2$. Let $0 < \delta < 1$. Consider a k -LIN $_q$ instance $\mathcal{I}$ of dimension n and m constraints, $m \geq \frac{l \ln(qn)}{\delta^2} \cdot \left(\frac{\alpha q n}{l}\right)^{k/2}$ for an absolute constant $\alpha > 0$. Let $\Delta = \frac{m\theta^{(n,l,k)}}{\binom{n}{l} q^l}$. Then for $\mathcal{I} \sim \mathcal{R}_{n,k}(m)$ we get $\|\mathcal{K}_l(\mathcal{I})\| \leq \delta\Delta$ except with probability at most $n^{-\Omega(l)}$.*

Corollary 5.11. *Let k, n, l, q be positive integers such that $k \leq l \leq n$. Let $0 < \delta < 1$. Consider a k -LIN $_q$ instance $\mathcal{I}$ of dimension n and m constraints, $m \geq \frac{l \ln(qn)}{\delta^2} \cdot \left(\frac{\alpha n}{l}\right)^k$ for an absolute constant $\alpha > 0$. Let $\Delta = \frac{m\theta^{(n,l,k)}}{\binom{n}{l} q^l}$. Then for $\mathcal{I} \sim \mathcal{R}_{n,k}(m)$ we get that $\|\mathcal{K}_l(\mathcal{I})\| \leq \delta\Delta$ except with probability at most $n^{-\Omega(l)}$.*

5.3 A lower bound for the planted instance

Recall that the only difference in a planted instance from a random instance of k -LIN $_q$ is that the ‘right-hand side’ $b_i \in \Omega_q$ of a constraint (S_i, b_i) is chosen from a non-uniform distribution.

The heart of our analysis will still be the Matrix Bernstein inequality, but with a slightly different set of matrices. This is because Lemma 5.7 ($\mathbb{E}[\mathbf{X}_i] = \mathbf{O}$) will no longer hold for the same definition of $\mathbf{X}_i$ when $\mathcal{I} \sim \mathcal{P}_{n,k}^z(m, \mathcal{D}_\rho)$. Let us calculate $\mathbb{E}[\mathbf{X}_i]$.

Notation 5.2. *Let $S \in \binom{[n]}{k} \times [q]^k$ and $\mathbf{z} \in \Omega_q^n$. Let $[N] := \binom{[n]}{l} \times [q]^l$. We define a $[N] \times [N]$ matrix $A_{\mathbf{z},S}$ as follows:*

$$A_{\mathbf{z},S}(T_1, T_2) := \begin{cases} \mathbf{z}^S & \text{if } \mathcal{S}(T_1, T_2) = S \\ (\mathbf{z}^S)^* & \text{if } \mathcal{S}(T_2, T_1) = S \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

Lemma 5.12. *Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]} \sim \mathcal{P}_{n,k}^z(m, \mathcal{D}_\rho)$ and $\mathbf{X}_i$ ’s be the same as defined in (6). Then $\mathbb{E}[\mathbf{X}_i] = \rho A_{\mathbf{z},S_i}$.*

Proof. This follows from the fact that $b_i = \eta_i(\mathbf{z}^{S_i})$ and $\mathbb{E}[\eta_i] = \mathbb{E}[\eta_i^*] = \rho$. $\square$

Notice that $\sum_{i \in [m]} A_{\mathbf{z}, S_i}$ is like a signed adjacency matrix of an appropriate graph. Analogous to the spectral norm of an adjacency matrix being lower bounded by the average degree of the graph, we will set a lower bound for the spectral norm of $\sum_{i \in [m]} A_{\mathbf{z}, S_i}$ which will be useful later.

Proposition 5.13. *Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]} \sim \mathcal{P}_{n,k}^{\mathbf{z}}(m, \mathcal{D}_\rho)$, $[N] = \binom{[n]}{l} \times [q]^l$ and $\Delta = \frac{m\theta^{(n,l,k)}}{N}$. Then $\left\| \sum_{i \in [m]} A_{\mathbf{z}, S_i} \right\| \geq \Delta$*

Proof.

$$\left\| \sum_{i \in [m]} A_{\mathbf{z}, S_i} \right\| = \sup_{\mathbf{x} \in \mathbb{C}^N} \frac{\mathbf{x}^\dagger \left(\sum_{i \in [m]} A_{\mathbf{z}, S_i} \right) \mathbf{x}}{\mathbf{x}^\dagger \mathbf{x}} \quad (11)$$

Let us consider the quadratic form in (11) for a specific $\mathbf{x} \in \mathbb{C}^N$. Define $\mathbf{z}^{\odot l} \in \mathbb{C}^N$ as follows: For $T \in [N]$, $\mathbf{z}_T^{\odot l} := (\mathbf{z}^T)^*$. It can be seen that for (T_1, T_2) satisfying $\mathcal{S}(T_1, T_2) = S$ for any set S , we have $(\mathbf{z}_{T_1}^{\odot l})^{-1} \mathbf{z}_{T_2}^{\odot l} = (\mathbf{z}^S)^*$

With this, we have

$$\begin{aligned} \left\| \sum_{i \in [m]} A_{\mathbf{z}, S_i} \right\| &\geq \frac{\left(\sum_{i \in [m]} (\mathbf{z}^{\odot l})^\dagger \cdot A_{\mathbf{z}, S_i} \cdot \mathbf{z}^{\odot l} \right)}{(\mathbf{z}^{\odot l})^\dagger \mathbf{z}^{\odot l}} \\ &= \frac{\sum_{i \in [m]} \sum_{(T_1, T_2)} (\mathbf{z}_{T_1}^{\odot l})^{-1} \cdot A_{\mathbf{z}, S_i}(T_1, T_2) \cdot \mathbf{z}_{T_2}^{\odot l}}{N} \\ &= \frac{\sum_{i \in [m]} (\# \text{ of non-zero entries in } A_{\mathbf{z}, S_i})}{N} \\ &= \frac{m\theta^{(n,l,k)}}{N} = \Delta \end{aligned}$$

$\square$

Let us now define $\widetilde{\mathbf{X}}_i := \mathbf{X}_i - \mathbb{E}[\mathbf{X}_i] = \mathbf{X}_i - \rho A_{\mathbf{z}, S_i}$. By definition, $\mathbb{E}[\widetilde{\mathbf{X}}_i] = 0$. Let us also obtain a bound on its spectral norm.

Lemma 5.14. *For $\widetilde{\mathbf{X}}_i$ defined as above, $\left\| \widetilde{\mathbf{X}}_i \right\| \leq 2(1 + \rho)$.*

Proof. $\left\| \widetilde{\mathbf{X}}_i \right\| \leq \left\| \mathbf{X}_i \right\| + \rho \left\| A_{\mathbf{z}, S_i} \right\| \leq 2(1 + \rho)$ where the second inequality can be obtained by an analogous argument for $A_{\mathbf{z}, S_i}$ as in Lemma 5.2. $\square$

To apply the Matrix Bernstein inequality, we are left to compute

$$\sigma^2 := \left\| \sum_{i \in [m]} \mathbb{E} \left(\widetilde{\mathbf{X}}_i^2 \right) \right\|$$

$$\sigma^2 = \left\| \sum_{i \in [m]} \mathbb{E} \left((\mathbf{X}_i - \rho A_{\mathbf{z}, S_i})^2 \right) \right\| = \left\| \sum_{i \in [m]} (\mathbb{E}(\mathbf{X}_i^2) - \rho^2 A_{\mathbf{z}, S_i}^2) \right\| \quad (12)$$

$$\leq \left\| \sum_{i \in [m]} \mathbb{E}(\mathbf{X}_i^2) \right\| + \rho^2 \left\| \sum_{i \in [m]} A_{\mathbf{z}, S_i}^2 \right\| \quad (13)$$

$$\leq 2(1 + \rho^2) D(\mathcal{K}_l(\mathcal{I})) \quad (\text{using Lemma 5.3 and same idea for } A_{\mathbf{z}, S_i}) \quad (14)$$

Like the random case, we need to choose an appropriate value of t to limit the probability in Theorem 5.6 to be essentially an inverse polynomial, say $N^{-\epsilon}$ for some choice of $\epsilon > 0$. Thus we must satisfy

$$N \cdot \exp\left(\frac{-t^2/2}{2(1 + \rho^2) D(\mathcal{K}_l(\mathcal{I})) + 2(1 + \rho)t/3}\right) \leq N^{-\epsilon} \quad (15)$$

It can be seen that (15) is satisfied for (denote $Z = (1 + \epsilon) \ln N$ in below equation)

$$t \geq \frac{2}{3} \left((1 + \rho)Z + \sqrt{((1 + \rho)Z)^2 + 9(1 + \rho^2)D(\mathcal{K}_l(\mathcal{I}))Z} \right) \quad (16)$$

Like the random case, the next step would have been to bound $D(\mathcal{K}_l(\mathcal{I}))$. Fortunately, we can again assume $D(\mathcal{K}_l(\mathcal{I})) \leq (1 + \gamma)\Delta$ from Proposition 5.5; even when $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]} \sim \mathcal{P}_{n,k}^{\mathbf{z}}(m, \mathcal{D}_\rho)$, the constraint sets $\{S_i\}_{i \in [m]}$ are chosen uniformly at random from $\binom{[n]}{k} \times [q]^l$.

So, let us now proceed to get an upper bound on $\left\| \sum_{i \in [m]} \widetilde{\mathbf{X}}_i \right\|$ with respect to Δ . Like before, we would like to choose Δ so that we get $\left\| \sum_{i \in [m]} \widetilde{\mathbf{X}}_i \right\| \leq \gamma\Delta$. Observe that in equation (16) we can impose

$$\Delta \geq \frac{12(1 + \rho^2)(1 + \gamma) + 4\gamma(1 + \rho)}{3\gamma^2} (1 + \epsilon) \ln N$$

to get this desired bound for all $\gamma > 0$. Again, the failure probability of Proposition 5.5 is bounded by $N^{-\epsilon}$ for this choice.

We now have $\left\| \sum_{i \in [m]} \widetilde{\mathbf{X}}_i \right\| \leq \gamma\Delta$. How do we proceed to get a lower bound for $\left\| \sum_{i \in [m]} \mathbf{X}_i \right\| = \|\mathcal{K}_l(\mathcal{I})\|$?

$$\begin{aligned} \|\mathcal{K}_l(\mathcal{I})\| &= \left\| \sum_{i \in [m]} \mathbf{X}_i \right\| \geq \left\| \sum_{i \in [m]} \mathbb{E}[\mathbf{X}_i] \right\| - \left\| \sum_{i \in [m]} \widetilde{\mathbf{X}}_i \right\| \\ &\geq \rho \left\| \sum_{i \in [m]} A_{\mathbf{z}, S_i} \right\| - \gamma\Delta \quad (\text{from Lemma 5.12}) \\ &\geq \rho\Delta - \gamma\Delta \quad (\text{from Proposition 5.13}) \end{aligned}$$

Theorem 5.15. Consider a k -LIN_q instance $\mathcal{I}$, let

$$\Delta = \frac{m\theta^{(n,l,k)}}{N} \geq \frac{12(1 + \rho^2)(1 + \gamma) + 4\gamma(1 + \rho)}{3\gamma^2} (1 + \epsilon) \ln N$$

(where symbols have the usual meaning). Then for $\mathcal{I} \sim \mathcal{P}_{n,k}^{\mathbf{z}}(m, \mathcal{D}_\rho)$, we get that $\|\mathcal{K}_l(\mathcal{I})\| \geq (\rho - \gamma)\Delta$ except with probability at most $2N^{-\epsilon}$.

Finally let us restate this result in terms of the parameters of $\mathcal{I}$. Once again we will use the lower bounds of $\theta^{(n,l,k)}$ from (3) and (4); the better result can be used depending on whether $q = O(n/l)$ or $q = \omega(n/l)$.

Corollary 5.16. *Let k, n, l, q be positive integers such that $k/2 \leq l \leq n - k/2$. Let $0 < \gamma < 1$. Consider a k -LIN $_q$ instance $\mathcal{I}$ of dimension n and m constraints, $m \geq \frac{l \ln(qn)}{\gamma^2} \cdot \left(\frac{\alpha q n}{l}\right)^{k/2}$ where $\alpha > 0$ is an absolute constant. Let $\Delta = \frac{m \theta^{(n,l,k)}}{\binom{n}{l} q^l}$. Then for $\mathcal{I} \sim \mathcal{P}_{n,k}^z(m, \mathcal{D}_\rho)$ for an unknown $z \in \Omega_q^n$, we get that $\|\mathcal{K}_l(\mathcal{I})\| \geq (\rho - \gamma)\Delta$ except with probability at most $n^{-\Omega(l)}$.*

Corollary 5.17. *Let k, n, l, q be positive integers such that $k \leq l \leq n$. Let $0 < \gamma < 1$. Consider a k -LIN $_q$ instance $\mathcal{I}$ of dimension n and m constraints, $m \geq \frac{l \ln(qn)}{\gamma^2} \cdot \left(\frac{\alpha n}{l}\right)^k$ where $\alpha > 0$ is an absolute constant. Let $\Delta = \frac{m \theta^{(n,l,k)}}{\binom{n}{l} q^l}$. Then for $\mathcal{I} \sim \mathcal{P}_{n,k}^z(m, \mathcal{D}_\rho)$ for an unknown $z \in \Omega_q^n$, we get that $\|\mathcal{K}_l(\mathcal{I})\| \geq (\rho - \gamma)\Delta$ except with probability at most $n^{-\Omega(l)}$.*

5.4 Distinguishing between a planted and a random instance

Corollaries 5.10, 5.11 and Corollaries 5.16, 5.17 together suggest that we can solve the decisional k -LIN $_q$ problem with high probability if we had the ability to approximate the maximum eigenvalue of $\mathcal{K}_l(\mathcal{I})$. For this, we need to ensure that the thresholds established in these corollaries are separated. In particular, we must have $(\rho - \gamma)\Delta > \delta\Delta$, which can be easily achieved by setting, say, $\gamma = \delta = \rho/4$.

We can then use a standard eigenvalue estimation technique, say the Power Iteration method, to approximate $\|\mathcal{K}_l(\mathcal{I})\|$. Note that $\mathcal{K}_l(\mathcal{I})$ is a $N \times N$ matrix for $N = \binom{n}{l} q^l$. Moreover, it is a sparse matrix since there are only Δ non-zero entries, where $\Delta = O(\ln N)$ suffices for our purpose. Thus, the Power Iteration takes $\tilde{O}(N) = \tilde{O}\left(\binom{n}{l} q^l\right)$ time to converge.

Corollary 5.18. *Let k, n, l, q be positive integers such that $k/2 \leq l \leq n - k/2$. Consider a k -LIN $_q$ instance $\mathcal{I}$ of dimension n and m constraints which is either sampled from $\mathcal{R}_{n,k}(m)$ or $\mathcal{P}_{n,k}^z(m, \mathcal{D}_\rho)$ for a given $\rho \in (0, 1]$, distribution $\mathcal{D}_\rho$ and an unknown $z \in \Omega_q^n$. Let $m \geq \frac{l \ln(qn)}{\rho^2} \cdot \left(\frac{\alpha q n}{l}\right)^{k/2}$ where $\alpha > 0$ is an absolute constant. Then we can determine the distribution of $\mathcal{I}$ in $\tilde{O}\left(\binom{n}{l} q^l\right)$ time except with probability at most $n^{-\Omega(l)}$.*

Corollary 5.19. *Let k, n, l, q be positive integers such that $k \leq l \leq n$. Consider a k -LIN $_q$ instance $\mathcal{I}$ of dimension n and m constraints which is either sampled from $\mathcal{R}_{n,k}(m)$ or $\mathcal{P}_{n,k}^z(m, \mathcal{D}_\rho)$ for a given $\rho \in (0, 1)$, distribution $\mathcal{D}_\rho$ and an unknown $z \in \Omega_q^n$. Let $m \geq \frac{l \ln(qn)}{\rho^2} \cdot \left(\frac{\alpha n}{l}\right)^k$ where $\alpha > 0$ is an absolute constant. Then we can determine the distribution of $\mathcal{I}$ in $\tilde{O}\left(\binom{n}{l} q^l\right)$ time except with probability at most $n^{-\Omega(l)}$.*

An important aspect of this distinguisher is that it works for any (nice enough) noise distribution of the planted instance. Let e be a random variable for the noise distribution. The sample complexity required by the algorithm is inversely proportional to ρ^2 where $\rho = \mathbb{E}[e^{2\pi i e/q}]$ and $\rho \in (0, 1]$. The sample and time complexities do not depend on any other property of the distribution. Moreover, ρ is sufficiently large (inverse polynomial in n) if the distribution is centered and sufficiently concentrated around 0, in which case the required sample complexity will be good. As immediate consequences of Corollaries 5.18, 5.19 we have algorithms for solving k -sparse LWE and sparse LPN over polynomial modulus q , if we take the planted distributions $D_\rho = \mathcal{LWE}_\rho$, $D_\rho = \mathcal{LPN}_\rho$ respectively.

For sparse LWE, from Remark A.2, we get that $1/\rho^2 \leq e^{2\pi r^2/q^2}$. Since the width r for the LWE distribution is chosen to be less than q , this dependency will only contribute a constant term.

Theorem 5.20. *Let n, m, q, r be positive integers, $r < q/2$. Let $\alpha > 0$ be an absolute constant. There exists an algorithm that solves the decisional k -LWE $_{n,m,q,r}$ problem such that:*

- *Let $l \in \mathbb{N}$ be a parameter such that $k/2 \leq l \leq n - k/2$. Then provided that the number of samples $m \geq l \ln(qn) \cdot (\frac{\alpha q n}{l})^{k/2}$, the algorithm takes time $\tilde{O}(\binom{n}{l} q^l)$ and is correct, except with probability at most $n^{-\Omega(l)}$.*
- *Let $l \in \mathbb{N}$ be a parameter such that $k \leq l \leq n$. Then provided that the number of samples $m \geq l \ln(qn) \cdot (\frac{\alpha n}{l})^k$, the algorithm takes time $\tilde{O}(\binom{n}{l} q^l)$ time and is correct except with probability at most $n^{-\Omega(l)}$.*

For sparse LPN, $\rho = 1 - \frac{q\mu}{q-1}$ from Definition 3.6. For $\mu \rightarrow \frac{q-1}{q}$, the planted distribution approaches the random distribution. If $\mu = \frac{q-1}{q} - \frac{1}{\text{poly}(n)}$, we have $\rho = 1/\text{poly}(n)$, and this only increases the sample complexity by a polynomial factor.

Theorem 5.21. *Let n, m, q be positive integers, $0 \leq \mu \leq \frac{q-1}{q} - \frac{1}{n^d}$ for some constant $d > 0$. Let $\alpha > 0$ be an absolute constant. There exists an algorithm that solves the decisional k -LPN $_{n,m,q,\mu}$ problem such that:*

- *Let $l \in \mathbb{N}$ be a parameter such that $k/2 \leq l \leq n - k/2$. Then provided that the number of samples $m \geq n^{2d} (l \ln(qn)) \cdot (\frac{\alpha q n}{l})^{k/2}$, the algorithm takes time $\tilde{O}(\binom{n}{l} q^l)$ time and is correct except with probability at most $n^{-\Omega(l)}$.*
- *Let $l \in \mathbb{N}$ be a parameter such that $k \leq l \leq n$. Then provided that the number of samples $m \geq n^{2d} (l \ln(qn)) \cdot (\frac{\alpha n}{l})^k$, the algorithm takes time $\tilde{O}(\binom{n}{l} q^l)$ time and is correct except with probability at most $n^{-\Omega(l)}$.*

5.5 A quantum speedup

In essence, to solve decisional k -LIN $_q$, the Kikuchi method performs estimation of the spectral norm of a sparse Hermitian matrix in a space of dimension $N = \binom{n}{l} q^l$. For any known classical method, such as the power iteration, eigenvalue estimation requires matrix multiplications, so it is unlikely for the time complexity to go below $\tilde{\Omega}(N)$.

However, with quantum algorithmic techniques, we can do better. We can use the same idea as given by Schmidhuber et. al. in [SOKB25] for $q = 2$: Using a *guiding state* created from the k -LIN $_q$ instance, we are able to determine whether the eigenspace corresponding to eigenvalues $\geq \lambda$ for some threshold $\lambda > 0$ is non-empty. This enjoys a speedup on Power iteration which can be from quadratic to quartic depending on the choice of parameter l . We elaborate on this intuition in Appendix B. For even more details, the reader may refer to Appendix C of [SOKB25].

6 Distinguishing by q -ary covers via Kikuchi Graphs

In this section, we will show a method to distinguish the planted vs random k -LIN $_q$ instances using a combinatorial structure in the hypergraph associated with the instance, which we call a q -ary cover. This is a generalization of *even cover* in k -XOR instances which has been studied in several works (eg. [FKO06; Fei08; GKM22]). Throughout this section, we assume that q is prime.

Unlike the spectral method attack, we are unable to analyze the effectiveness of this attack for any noise distribution in the planted instance $\mathcal{D}_\rho$. Instead we now focus on two kinds of noise distributions on $\mathbb{Z}_q$. The first kind is where the error is bounded in a small region $[-r, r]$ for $r \ll q$ such as the discrete Gaussian with width parameter r (the most popular LWE noise distribution). The second is the distribution used to define LPN for higher moduli; concentrated on 0 and uniform on non-zero points of $\mathbb{Z}_q$.

For the first kind, we can distinguish in a relatively simple way once we obtain a q -ary cover. For the LPN distribution, our technique is very similar to that of [GHOS26] where they find many even covers to distinguish a planted vs random k -XOR instance.

For both distributions, the q -ary cover method is also able to distinguish *worst-case* instances (unlike the spectral method), i.e. when the k -sparse constraints are chosen arbitrarily instead of uniformly.

6.1 q -ary covers

Let us begin by defining a q -ary cover.

Definition 6.1. (*q -ary cover*) Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be a k -LIN $_q$ instance on n variables. A cover $\mathbf{c}$ is a multiset containing constraints in $\mathcal{I}$, which can be viewed as a vector $\mathbf{c} \in (\mathbb{N} \cup \{0\})^m$ whose i^{th} entry c_i depicts the number of times the constraint (S_i, b_i) occurs in the cover. $\mathbf{c}$ is said to be a q -ary cover of $\mathcal{I}$ if for every $j \in [n]$

$$\sum_{i \in [m]} (\text{coeff}_{S_i}(j) \cdot c_i) \cong 0 \pmod{q} \quad (17)$$

Remark 6.2. In the rest of the discussion, we view $\mathbf{c}$ of Definition 6.1 as a vector in $\mathbb{Z}_q^m$ as $\mathbf{c} \pmod{q}$ is a q -ary cover iff $\mathbf{c}$ is.

Remark 6.3. A key difference in our definition and the even cover definition for $q = 2$ is that we allow multiple copies of a constraint in $\mathbf{c}$. Note that the multiset definition does not make a difference in the $q = 2$ case since the count of a constraint and the count modulo 2 (present or absent) has the same effect on (17).

We wish to utilize the structure of q -ary cover to distinguish between planted and random instances. For a non-trivial q -ary cover $\mathbf{c}$ ($\mathbf{c} \neq \mathbf{0}$), we will leverage the fact that the distribution of the right-hand sides of the constraints in a q -ary cover (considered with multiplicities) differs significantly between the two cases.

Let us first describe how to find q -ary covers. For this, we relate a q -ary cover of $\mathcal{I}$ with a closed random walk in its Kikuchi graph $\mathcal{K}_l(\mathcal{I})$. In the context of this section, a walk in a directed graph $G = (V, E)$ is only in the direction of the edges; i.e. $W = (w_0, w_1, \dots, w_T)$ is a walk only if $(w_{t-1}, w_t) \in E \forall t \in [T]$.

Proposition 6.1. Suppose $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ is a k -LIN $_q$ instance. Let $W = (w_0, w_1, \dots, w_T = w_0)$ be a closed walk in $\mathcal{K}_l(\mathcal{I})$. Construct $\mathbf{c} = (c_i)_{i \in [m]}$ as follows (c_i 's are initialized to 0):

For $r \in [T]$, if $\mathcal{S}(w_{r-1}, w_r) = S_i$ for some $i \in [m]$ then add 1 to c_i , else if $\mathcal{S}(w_r, w_{r-1}) = S_i$ for some $i \in [m]$ then add $q - 1$ to c_i .

Then $\mathbf{c}$ is a q -ary cover of $\mathcal{I}$.

Proof. For $r \in [T]$, denote $S'_r := \mathcal{S}(w_{r-1}, w_r)$ (we emphasize that S'_r may coincide either with some S_i or its ‘conjugate’). First, observe that for any $j \in [n]$

$$\sum_{i \in [m]} \text{coeff}_{S_i}(j) \cdot c_i \cong \sum_{r \in [T]} \text{coeff}_{S'_r}(j) \pmod{q}$$

We therefore count the coefficients of j over $\{S'_r\}_{r \in [T]}$. Recall that for each $r \in [T]$, $w_r \in \binom{[n]}{l} \times [q]^l$. Then, by the definition of the operator $\mathcal{S}$, we have $\text{coeff}_{S'_r}(j) = \text{coeff}_{w_r}(j) - \text{coeff}_{w_{r-1}}(j)$. Thus, summing the coefficients of j over every S'_r (remember $w_T = w_0$) gives 0 mod q . $\square$

We have reduced the problem of finding q -ary covers in $\mathcal{I}$ to finding closed walks in $\mathcal{K}_l(\mathcal{I})$. However, we need to ensure that the q -ary cover corresponding to a closed walk is not trivial. We will argue analogously to [GHOS26] that a closed walk obtained by a random process will yield a non-trivial q -ary cover with high probability.

6.1.1 Good Closed Walks.

We will now develop algorithms to find distinct ‘good’ closed walks in any directed graph, and we will later use them for the Kikuchi graph.

Let us first establish some useful graph-theoretic tools.

Definition 6.4. For a directed graph $G = (V, E)$, let ν denote the out-degree distribution over V given by $\nu(v) \propto \text{out-deg}(v)$ for all $v \in V$.

Fact 6.2. If G has average out-degree Δ , then for any $0 < \beta \leq 1$, it holds that

$$\Pr_{v \sim \nu}[\text{out-deg}(v) < \beta \Delta] < \beta.$$

Proof. Let $S := \{v : \text{out-deg}(v) < \beta \Delta\}$. Then

$$\Pr_{v \sim \nu}[\text{out-deg}(v) < \beta \Delta] = \frac{\sum_{v \in S} \text{out-deg}(v)}{|E|} < \frac{\sum_{v \in S} \beta \Delta}{|E|} = \frac{\beta \Delta |S|}{|E|} \leq \frac{\beta \Delta |V|}{|E|} = \beta.$$

$\square$

Definition 6.5. The random walk distribution of length T from w_0 , denoted $\text{walk}_{w_0}^T$, is a distribution over walks of length T with initial vertex w_0 defined as follows: To sample $W = (w_0, w_1, \dots, w_T) \sim \text{walk}_{w_0}^T$, iteratively sample w_{t+1} to be a random neighbor of w_t for $1 \leq t < T$.

The following fact holds for special directed graphs, which satisfy $\text{in-deg}(v) = \text{out-deg}(v) \forall v \in V$. The Kikuchi graph satisfies this property.⁹

Fact 6.3. Let $G = (V, E)$ be a directed graph such that for every vertex $v \in V$, $\text{in-deg}(v) = \text{out-deg}(v)$. Let $T \in \mathbb{N}$. If $w_0 \sim \nu$ and $W = (w_0, w_1, \dots, w_T) \sim \text{walk}_{w_0}^T$, then the marginal distribution of w_t is ν (for every $1 \leq t \leq T$).

⁹such a graph is equivalent to an undirected graph for all practical purposes. However we are associating signs with each directed edge of the Kikuchi graph, and edge pairs (v, w) and (w, v) have different signs, so we prefer to talk in terms of directed graphs.

Proof. Let us assume by inductive hypothesis that $w_{t-1} \sim \nu$ for some $t > 0$.

$$\begin{aligned} \Pr[w_t] &= \sum_{(w_{t-1}, w_t) \in E} \Pr[w_t | w_{t-1}] \Pr[w_{t-1}] \\ &= \sum_{(w_{t-1}, w_t) \in E} \frac{1}{\text{out-deg}(w_{t-1})} \frac{\text{out-deg}(w_{t-1})}{|E|} = \frac{\text{in-deg}(w_t)}{|E|} = \frac{\text{out-deg}(w_t)}{|E|} \end{aligned}$$

□

We now define a ‘goodness’ property of a walk in a directed graph G . Intuitively, a walk is good if it does not have too many vertices of low degree.

Definition 6.6. Let $\beta \in (0, 1]$. Let $G = (V, E)$ be a directed graph. Let $\Delta = \frac{|E|}{|V|}$ be the average out-degree of the graph. We say that a vertex $v \in V$ is β -bad if $\text{out-deg}(v) < \beta\Delta$. For a directed walk $W = (w_0, \dots, w_T)$ in G , let $B_\beta(W) := |\{0 \leq i < T : w_i \text{ is } \beta\text{-bad}\}|$. For $\gamma \geq 1$, we say that W is (γ, β) -good if $B_\beta(W) < \gamma\beta T$.

Lemma 6.4. Let $G = (V, E)$ be a directed graph such that for every vertex $v \in V$, $\text{in-deg}(v) = \text{out-deg}(v)$. Suppose $w_0 \sim \nu$ and $W \sim \text{walk}_{w_0}^T$, then

$$\Pr \left[W \text{ is } \left(\frac{1}{1-\varepsilon}, \beta \right)\text{-good} \right] > \varepsilon$$

Proof. By Fact 6.3 and Fact 6.2, we get that $\Pr[w_t \text{ is } \beta\text{-bad}] < \beta$ for each $w_t \in W$, where the probability is over both $w_0 \sim \nu$ and $W \sim \text{walk}_{w_0}^T$. Thus

$$\mathbb{E}_{w_0 \sim \nu, W \sim \text{walk}_{w_0}^T} [B_\beta(W)] < \beta T$$

By Markov’s inequality, $\Pr[W \text{ is } (\gamma, \beta)\text{-good}] = \Pr[B_\beta(W) < \gamma\beta T] > 1 - \frac{1}{\gamma} = \varepsilon$ for $\gamma = \frac{1}{1-\varepsilon}$. □

We are now ready to present the following algorithm (analogous to Algorithm 1 in [GHOS26]) to obtain a good closed walk on a directed graph G . For two walks W_i, W_j with the end vertex of W_i coinciding with the start vertex of W_j , we use $W_i \circ W_j$ to denote the concatenated walk.

Algorithm 1 FIND-GOOD-CLOSED-WALK($G, w_0, T, \beta, \varepsilon$)

Require: Description of directed graph $G = (V, E)$ ¹⁰ wherein every edge $e = (u, v)$ has a corresponding reverse edge $e^{\text{rev}} = (v, u)$, $|V| = N$, start vertex $w_0 \in V$, walk length $T \in \mathbb{N}$, goodness parameters $\beta \in (0, 1], \varepsilon \in (0, 1)$

- 1: Draw $W_1, \dots, W_L \sim \text{walk}_{w_0}^T$ independently, where $L := \lceil C_1 \sqrt{N \log N} \rceil$
 - 2: **if** there exist $i \neq j \in [L]$ such that W_i, W_j are both $\left(\frac{1}{1-\varepsilon}, \beta\right)$ -good and end at the same vertex **then**
 - 3: **return** $W_i \circ W_j^{\text{rev}}$
 - 4: **end if**
 - 5: **return** FAIL
-

Before proving the correctness of Algorithm 1, let us state a standard fact, which is essentially the famous McLaurin inequality (see eg., §2.22 of [HLP52]).

¹⁰we don’t need to store the graph, we just need enough description to compute neighbors of any vertex in the graph

Fact 6.5. For every finite set S , distribution $\mathcal{D}$ on S , and $k \in \mathbb{N}$,

$$\Pr_{X_1, \dots, X_k \stackrel{iid}{\sim} \mathcal{D}} [\exists \text{ collision among } X_1, \dots, X_k] \geq \Pr_{X_1, \dots, X_k \stackrel{iid}{\sim} \text{Unif}(S)} [\exists \text{ collision among } X_1, \dots, X_k]$$

We now prove that Algorithm 1 will successfully return a closed walk with both halves good; the proof resembles that of Lemma 18 in [GHOS26].

Lemma 6.6. Let $G = (V, E)$ be a directed graph such that every edge $e = (u, v)$ has a corresponding reverse edge $e^{\text{rev}} = (v, u)$. Fix $\beta \in (0, 1], \varepsilon \in (0, 1)$. Suppose $w_0 \sim \nu$. FIND-GOOD-CLOSED-WALK($G, w_0, T, \beta, \varepsilon$) successfully returns a closed walk $W = W_i \circ W_j^{\text{rev}}$ where both W_i, W_j are of length T and are $\left(\frac{1}{1-\varepsilon}, \beta\right)$ -good except with probability $1/\text{poly}(N)$.

Proof. Firstly, by Lemma 6.4 we get that $\Pr \left[W_i \text{ is } \left(\frac{1}{1-\varepsilon}, \beta\right)\text{-good} \right] > \varepsilon$ for all $i \in [L]$. By Chernoff bound, we get that there are at least $(\varepsilon/2)L$ walks which are $\left(\frac{1}{1-\varepsilon}, \beta\right)$ -good except with probability at most $\exp(-\varepsilon L/8)$. Conditioned on this event, let $\mathcal{D}$ be the distribution of final vertices of the walks $W_1, \dots, W_L$. By Fact 6.5 and an application of the Birthday paradox, we get that probability of no two walks ending at the same vertex is at most $\exp\left(-\frac{(\varepsilon L/2)((\varepsilon L/2)-1)}{2N}\right) \leq \exp\left(-\frac{\varepsilon C_1^2 \log N}{4}\right)$ assuming large enough N . Adding both the failure probability gives us a failure probability of inverse polynomial in N . $\square$

We can use Algorithm 1 on the Kikuchi Graph to find a closed walk which will correspond to a q -ary cover. As discussed, the second part of the task is to ensure ‘non-trivial’ covers.

Recall from Proposition 6.1 that we could extract a q -ary cover for $\mathcal{I}$ from a closed walk on $\mathcal{K}_l(\mathcal{I})$ by looking at the edge labels of the walk, which correspond to the constraints of $\mathcal{I}$. For our analysis further, we’ll need to introduce these edge labels/colors in our directed graph model. Let us define an edge-coloring for a directed graph:

Definition 6.7. Let $G = (V, E)$ be a directed graph and $\Gamma : E \rightarrow [m]$ be a function. For $c \geq 1$, Γ is a c -edge-coloring if for every vertex $v \in V$ and every color $\gamma \in [m]$, we have

$$|\{w : e = (v, w) \in E, \Gamma(e) = \gamma\}| \leq c, \quad |\{w : e = (w, v) \in E, \Gamma(e) = \gamma\}| \leq c$$

We then say that $G = (V, E, \Gamma)$ is a c -edge colored directed graph.

Let $G = (V, E, \Gamma)$ be a c -edge colored directed graph with average degree Δ . We will now show that for a closed walk with both halves good, it will have a large number of edges whose color is seen unique to them in the walk.

Definition 6.8. Let $G = (V, E, \Gamma)$ be a c -edge colored directed graph with average degree Δ . Let W_1, W_2 be two walks in G , and denote $\mathcal{E} = \mathcal{E}(W_1, W_2) = (e_1, \dots, e_T, e_{T+1}, \dots, e_{2T})$ for the sequence of edges in W_1 followed by those in W_2 . Let $\text{Single}(W_1, W_2)$ be the set of singleton colors in $\mathcal{E}$, i.e. the set of colors which occur only once in $(\Gamma(e_i))_{i \in [2T]}$.

We now prove the following lemma analogously as Lemma 22 of [GHOS26].

Lemma 6.7. Let $G = (V, E, \Gamma)$ be a c -edge colored directed graph with average degree Δ . Fix $\beta \in (0, 1], \gamma > 1$. Fix $w_0 \in V$ and an ‘avoidance’ set of colors $A \subseteq [m]$ such that $|A| \leq 2T$. Suppose we independently sample $W_1, W_2 \sim \text{walk}_{w_0}^T$ and define the event:

$\mathcal{C} : W_1, W_2 \text{ are } (\gamma, \beta)\text{-good and } \text{Single}(W_1, W_2) \subseteq A.$

Then we have $\Pr[\mathcal{C}] \leq (CT/\Delta)^{C'T}$ for some constants $C = C(\beta, c), C' = C'(\gamma, \beta).$

Proof. For $i \in [2T]$, let us call the i th time step *unusual* if:

1. w_{i-1} is not β -bad, yet
2. $\Gamma(e_i) \in \{\Gamma(e_1), \dots, \Gamma(e_{i-1})\} \cup A$

We begin by arguing that if the event $\mathcal{C}$ occurs, then there is a large number of unusual steps.

First, let us identify the steps that violate condition (1). Since W_1, W_2 are (γ, β) -good, by Definition 6.6 there are less than $2\gamma\beta T$ vertices in W_1, W_2 combined which are β -bad. This means condition (1) is violated for less than $2\gamma\beta T$ steps.

Now let us turn to the violations of condition (2). If $\mathcal{C}$ occurs then $\text{Single}(W_1, W_2) \subseteq A$. This means that if $\Gamma(e_i)$ is a color that appears only once, it cannot violate (2) since it must be present in A . Therefore $\Gamma(e_i)$ can violate (2) only if it appears more than once in the sequence. But since condition (2) is not violated at step i if $\Gamma(e_i)$ has already been encountered (i.e. it is in $\{\Gamma(e_1), \dots, \Gamma(e_{i-1})\}$), every color that appears more than once can cause only one violation of (2). Hence, such violations can appear at no more than half the total number of steps, i.e., at no more than $2T/2 = T$ steps.

Thus both conditions are met for at least $2T - T - 2\gamma\beta T$ steps, i.e. there are at least $(1 - 2\gamma\beta)T = C'T$ unusual steps.

Let $U(W_1, W_2) \in \{0, 1\}^{2T}$ denote the *unusualness* string, i.e. $U(W_1, W_2)_i = 1$ iff the i^{th} step is unusual. From above, we obtain

$$\Pr[\mathcal{C}] \leq \Pr[|U(W_1, W_2)| \geq C'T] = \sum_{u: |u| \geq C'T} \Pr[U(W_1, W_2) = u] \quad (18)$$

Let us now bound $\Pr[U(W_1, W_2) = u]$ for a fixed string $u \in \{0, 1\}^{2T}$. For the same, we will bound the probability:

$$\Pr[U(W_1, W_2)_j = u_j = 1 \mid U(W_1, W_2)_{<j} = u_{<j}]$$

Note that for $U(W_1, W_2)_j = 1$, condition 1. says that w_{j-1} must not be β -bad, i.e. $\text{out-deg}(w_{j-1}) \geq \beta\Delta$. Further, condition 2. says that

$$\Gamma(e_j) \in \{\Gamma(e_1), \dots, \Gamma(e_{j-1})\} \cup A$$

There are at most $j - 1 + |A|$ choices for $\Gamma(e_j)$. Since the graph G is c -edge colorable, we can have at most $c(j - 1 + |A|)$ outgoing edges from w_{j-1} that have colors in $\{\Gamma(e_1), \dots, \Gamma(e_{j-1})\} \cup A$. The edge e_j is picked uniformly from the set of outgoing edges of w_{j-1} , thus we get

$$\Pr[U(W_1, W_2)_j = u_j = 1 \mid U(W_1, W_2)_{<j} = u_{<j}] \quad (19)$$

$$\leq \Pr[\Gamma(e_j) \in \{\Gamma(e_1), \dots, \Gamma(e_{j-1})\} \cup A \mid w_{j-1} \text{ is not } \beta\text{-bad}] \quad (20)$$

$$\leq \frac{c(j + |A|)}{\beta\Delta} \leq \frac{4cT}{\beta\Delta} \quad (21)$$

Thus,

$$\Pr[U(W_1, W_2) = u] = \prod_{j \in [2T]} \Pr[U(W_1, W_2)_j = u_j \mid U(W_1, W_2)_{<j} = u_{<j}] \quad (22)$$

$$\leq \prod_{\substack{j \in [2T] \\ u_j = 1}} \Pr[U(W_1, W_2)_j = u_j = 1 \mid U(W_1, W_2)_{<j} = u_{<j}] \quad (23)$$

$$\leq \left(\frac{4cT}{\beta\Delta} \right)^{|u|} \quad (\text{using (21)}) \quad (24)$$

Finally, we use (24) in (18) to complete the proof:

$$\Pr[C] \leq \sum_{u: |u| \geq C'T} \Pr[U(W_1, W_2) = u] \leq 2^{2T} \left(\frac{4cT}{\beta\Delta} \right)^{C'T} = \left(\frac{CT}{\Delta} \right)^{C'T}$$

□

Corollary 6.8. Fix $\beta \in (0, 1], \varepsilon \in (0, 1)$, and let $\gamma = \frac{1}{1-\varepsilon}$. Let $G = (V, E, \Gamma : E \rightarrow [m])$ be a c -edge colored directed graph such that every edge $e = (u, v)$ has a corresponding reverse edge $e^{\text{rev}} = (v, u)$, and let its average degree be Δ . Let $A \subseteq [m]$ such that $|A| \leq 2T$. Further, suppose $(3/C) \ln N < T < \Delta/(cC')$, where $C(\beta, c), C'(\gamma, \beta)$ are the constants in Lemma 6.7. Suppose $w_0 \sim \nu$. FIND-GOOD-CLOSED-WALK($G, w_0, T, \beta, \varepsilon$) successfully returns a closed walk $W = W_1 \circ W_2$ such that $\text{Single}(W) \not\subseteq A$ except with probability $1/N$.

Proof. By Lemma 6.6, we get that FIND-GOOD-CLOSED-WALK($G, w_0, T, \beta, \varepsilon$) returns a closed walk W with both halves good except with probability $1/\text{poly}(N)$. The algorithm also samples $O(L^2) = O(N \log N)$ pairs of walks sampled independently from $\text{walk}_{w_0}^T$. Lemma 6.7 says that any such good pair W_1, W_2 satisfies $\text{Single}(W_1, W_2) \subseteq A$ with probability at most $1/N^3$ (putting conditions on T and Δ). By a union bound, any of the pairs can have $\text{Single}(W_1, W_2) \subseteq A$ with probability at most $O(N \log N/N^3) \ll 1/2N$. Thus the total failure probability is bounded by $1/N$. □

Algorithm 1 returns a closed walk which is ‘non-trivial’ in the sense that its set of singleton colors is not contained in any small fixed set A . As we shall see, this is a much stronger non-trivial condition than what we directly require from the Kikuchi graph. Nevertheless, this condition is also useful for finding many non-trivial closed walks (and, subsequently, many q -ary covers) that are distinct from each other. We now present the algorithm FIND-DISTINCT-NONTRIVIAL-CLOSED-WALKS (Algorithm 2) for finding many non-trivial closed walks.

Corollary 6.9. Let $G = (V, E, \Gamma : E \rightarrow [m])$ be a c -edge colored directed graph such that every edge $e = (u, v)$ has a corresponding reverse edge $e^{\text{rev}} = (v, u)$. Fix $\varepsilon \in (0, 0.1)$. Suppose that G has average degree Δ such that $(3/C) \ln N < T < \Delta/(cC')$, where $C(\beta, c), C'(\gamma, \beta)$ are the constants in Lemma 6.7. Then with probability at least $1 - \exp(-N^\varepsilon)$, FIND-DISTINCT-NONTRIVIAL-CLOSED-WALKS($G, \delta, T, \varepsilon$) returns a list $\mathcal{W}$ of $\lceil 10N^\varepsilon \rceil$ nontrivial closed walks of length $\leq 2T$, with $\text{Single}(W)$ distinct for each walk $W \in \mathcal{W}$.

Proof. From Corollary 6.8, the r^{th} iteration produces a closed walk W_r which satisfies $\text{Single}(W_r) \not\subseteq \Gamma(W_{r'})$ for some previous walk $W_{r'}$, except with probability $1/N$. By a union bound, $\text{Single}(W_r) \not\subseteq \Gamma(W_{r'})$ for every $r' < r$ except with probability r/N . Let X_r be the indicator variable for the success

Algorithm 2 FIND-DISTINCT-NONTRIVIAL-CLOSED-WALKS(G, T, ε)

Require: A c -edge colored directed graph $G = (V, E, \Gamma : E \rightarrow [m])$ wherein every edge $e = (u, v)$ has a corresponding reverse edge $e^{\text{rev}} = (v, u)$, failure parameter δ , walk length T , parameter ε .

```
1:  $\mathcal{W} := \emptyset$ 
2: for  $r = 1, \dots, R := 100N^\varepsilon$  do
3:   Fix constants  $\beta, \varepsilon'$  (say  $\beta = \varepsilon = 0.5$ )
4:   Run FIND-GOOD-CLOSED-WALK( $G, v, T, \beta, \varepsilon'$ ) for  $v \sim \nu$ ;
5:   if it produces a nontrivial closed walk  $W_r$  ( $\text{Single}(W_r) \neq \emptyset$ ), and  $\text{Single}(W_r) \neq \Gamma(W_r)$  for all  $r' < r$  then
6:     add  $W_r$  to  $\mathcal{W}$ ;
7:   end if
8: end for
9: return  $\mathcal{W}$ 
```

of this event, i.e. $\Pr[X_r = 1] \geq 1 - r/N \geq 1 - R/N$, where $R = 100N^\varepsilon$ as defined in Algorithm 2. We construct $Y_r = \mathbb{E}[\sum_{i=1}^R X_i \mid X_1, \dots, X_r]$ which is the Doob Martingale associated with the sum $\sum_{i=1}^R X_i$. Note that $Y_R = \sum_{i=1}^R X_i$ and $Y_0 = \mathbb{E}[\sum_{i=1}^R X_i]$. Since $\{Y_r\}_{r=0}^R$ is a martingale, and satisfies $Y_r - Y_{r-1} \leq 1$, we can apply Azuma-Hoeffding's inequality to it, getting

$$\Pr[Y_R \leq Y_0 - t] \leq \exp\left(-\frac{t^2}{2R}\right) \quad (25)$$

Noting that $Y_0 = \sum_{r \in [R]} \mathbb{E}[X_r] = R - R^2/N$, we get

$$\Pr[Y_R \leq 0.1R] \leq \exp(-0.4R) \quad (\text{using } t = 0.9R \text{ and ignoring } R^2/N = N^{2\varepsilon-1}).$$

□

We now relate the Kikuchi Graph $\mathcal{K}_l(\mathcal{I})$ with a c -edge colored directed graph. The following proposition immediately follows from the properties of Kikuchi graph (see Remark 4.12).

Proposition 6.10. *Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be a k -LIN $_q$ instance, and $\mathcal{K}_l(\mathcal{I})$ be its Kikuchi graph for a parameter $l \geq k/2$. $\mathcal{K}_l(\mathcal{I})$ satisfies the property that every edge $e = (u, v)$ has a corresponding reverse edge $e^{\text{rev}} = (v, u)$. Further, for every edge (v, w) satisfying $\mathcal{S}(v, w) = S_i$, let Γ map the edges $(v, w), (w, v)$ to color i . Then $\mathcal{K}_l(\mathcal{I})$ is a 2-edge colored directed graph.*

Recall that the only ‘trivial’ q -ary covers which do not serve our purpose are those which have count of every constraint as 0 mod q . Since Corollary 6.9 guarantees that with high probability, Algorithm 2 produces closed walks with a non-empty set of singleton colors, the q -ary cover associated with such a walk (see Proposition 6.1) is non-trivial.

We can combine above results to have an algorithm for a non-trivial q -ary cover for a k -LIN $_q$ instance $\mathcal{I}$.

Corollary 6.11. *Let k, n, l, q be positive integers such that $k/2 \leq l \leq n - k/2$. Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be a k -LIN $_q$ instance of dimension n and m constraints. Let $m \geq l \ln(qn) \cdot \left(\frac{\alpha q n}{l}\right)^{k/2}$ where $\alpha > 0$ is an absolute constant. There is an algorithm that runs in time $\tilde{O}\left(\sqrt{\binom{n}{l}} q^l (n/l)^k\right)$ and finds a non-trivial q -ary cover $\mathbf{c}$ of $\mathcal{I}$ having $\|\mathbf{c}\|_1 = O(l \log n)$ except with probability $n^{-\Omega(l)}$.*

The proof of Corollary 6.11 is similar to that of Corollary 6.12 below.

Corollary 6.12. *Let k, n, l, q be positive integers such that $k \leq l \leq n$. Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be a k -LIN $_q$ instance of dimension n and m constraints. Let $m \geq l \ln(qn) \cdot \left(\frac{\alpha n}{l}\right)^k$ where $\alpha > 0$ is an absolute constant. There is an algorithm that runs in time $\tilde{O}\left(\sqrt{\binom{n}{l} q^l (n/l)^k}\right)$ and finds a non-trivial q -ary cover $\mathbf{c}$ of $\mathcal{I}$ having $\|\mathbf{c}\|_1 = O(l \log n)$ except with probability $n^{-\Omega(l)}$.*

Proof. We can use Algorithm 1 on $\mathcal{K}_l(\mathcal{I})$ and a vertex w_0 sampled according to ν . The correctness of the algorithm follows immediately from Proposition 6.1, Corollary 6.8, Proposition 6.10. In particular, Proposition 6.1 also gives us that the size of the q -cover $\|\mathbf{c}\|_1$ is the same as the size of the closed walk $2T \leq O(\log N) = O(l \log n)$ of $\mathcal{K}_l(\mathcal{I})$. Let us now argue the time complexity.

In Algorithm 1, step 1 takes $O(L) = O(N^{1/2} \log N)$ iterations. It involves finding neighbors of vertices of the Kikuchi Graph $\mathcal{K}_l(\mathcal{I})$. Each vertex of the graph can be represented in $\tilde{O}(l)$ bits, and there are m possible neighbors (corr. to m constraints) for any vertex. Since we sample a walk of length T , step 1 takes $\tilde{O}(N^{1/2} m l T \log N) = \tilde{O}(m \sqrt{\binom{n}{l} q^l})$ time. (Here we substituted $T = \Theta(\log N)$ from above conditions.) Steps 2-4 involve $\tilde{O}(L)$ checks, each of which can be performed in $\tilde{O}(l \log L)$ time. So this time is $\tilde{O}(N^{1/2})$. Overall the algorithm takes $\tilde{O}(m \sqrt{\binom{n}{l} q^l})$ time.

Let us compute the minimum sample complexity m . Corollary 6.9 gives us $\Delta > T \geq \frac{C'' \log N}{\theta^{(n,l,k)}}$ for a constant C'' . The average degree Δ of $\mathcal{K}_l(\mathcal{I})$ is given by $\Delta = \frac{m \theta^{(n,l,k)}}{N}$. Thus $m \geq \frac{C'' N \log N}{\theta^{(n,l,k)}}$. Using a lower bound for $\theta^{(n,l,k)}$ from (4) gives us that $m \geq l \ln(qn) \left(\frac{\alpha n}{l}\right)^k$ suffices for an absolute constant $\alpha > 0$. Substituting this in above time complexity completes the proof. $\square$

We can also combine above results to have an algorithm for many distinct non-trivial q -ary covers for a k -LIN $_q$ instance $\mathcal{I}$.

Corollary 6.13. *Let k, n, l, q be positive integers such that $k/2 \leq l \leq n - k/2$. Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be a k -LIN $_q$ instance of dimension n and m constraints. Let $m \geq l \ln(qn) \cdot \left(\frac{\alpha q n}{l}\right)^{k/2}$ where $\alpha > 0$ is an absolute constant. There is an algorithm that runs in time $\tilde{O}\left(\left(\binom{n}{l} q^l\right)^{1/2+\varepsilon} (n/l)^k\right)$ and finds $10N^\varepsilon$ non-trivial q -ary covers of $\mathcal{I}$ of sizes $O(l \log n)$ except with probability $n^{-\Omega(\varepsilon l)}$.*

The proof of Corollary 6.13 is similar to that of Corollary 6.14 below.

Corollary 6.14. *Let k, n, l, q be positive integers such that $k \leq l \leq n$. Let $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ be a k -LIN $_q$ instance of dimension n and m constraints. Let $m \geq l \ln(qn) \cdot \left(\frac{\alpha n}{l}\right)^k$ where $\alpha > 0$ is an absolute constant. There is an algorithm that runs in time $\tilde{O}\left(\left(\binom{n}{l} q^l\right)^{1/2+\varepsilon} (n/l)^k\right)$ and finds $10N^\varepsilon$ non-trivial q -ary covers of $\mathcal{I}$ of sizes $O(l \log n)$ except with probability $n^{-\Omega(\varepsilon l)}$.*

Proof. We can use Algorithm 2 on $\mathcal{K}_l(\mathcal{I})$; the correctness of the algorithm follows immediately from Proposition 6.1, Corollary 6.9, Proposition 6.10. Let us now argue the time complexity.

Algorithm 2 calls Algorithm 1 $O(N^\varepsilon)$ times. This contributes to the dominating time for each such call, so the total time so far (using Corollary 6.12) is $\tilde{O}\left(N^\varepsilon m \sqrt{\binom{n}{l} q^l}\right) = \tilde{O}\left(m \left(\binom{n}{l} q^l\right)^{1/2+\varepsilon}\right)$.

We can substitute the sufficient lower bound for m ; $m \geq l \ln n \left(\frac{\alpha n}{l}\right)^k$ as calculated in above proof of Corollary 6.12 to finish the proof. $\square$

6.2 Solving sparse LWE

Let us see how a non-trivial q -ary cover helps in distinguishing planted vs random k -LIN $_q$ instances, when the planted noise distribution has a bounded error with high probability.

For this purpose, let us view a k -LIN $_q$ instance as a system of linear equations instead of product equations. A k -LIN $_q$ instance $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ corresponds to $\{(\mathbf{a}_i, \beta_i)\}_{i \in [m]}$ according to Definition 3.5. We can write this instance in matrix form: (A, β) where $A \in \mathbb{Z}_q^{m \times n}$, $\beta \in \mathbb{Z}_q^m$ has i^{th} rows as $(\mathbf{a}_i, \beta_i)$.

It can be seen that $\mathbf{c} \in \mathbb{Z}_q^m$ is a q -ary cover iff $\mathbf{c}^T A = \mathbf{0}$. Using this fact, we analyze the distribution $\mathbf{c}^T \beta$ to distinguish the two cases.

In the planted case, $|\mathbf{c}^T \beta| = |\mathbf{c}^T (A\mathbf{s} + \mathbf{e})| = |\mathbf{c}^T \mathbf{e}| \leq \|\mathbf{c}\|_1 \|\mathbf{e}\|_\infty$ where the last inequality is the Hölder's inequality.¹¹ From Corollaries 6.11, 6.12, we are able to obtain $\mathbf{c}$ satisfying $\|\mathbf{c}\|_1 \leq O(l \log n)$. Further, if we have a bound on the error vector $\|\mathbf{e}\|_\infty \leq B$ with high probability, then we get that $|\mathbf{c}^T \beta| \leq O(Bl \log n)$ with high probability.

In the random case, β is uniformly random in $\mathbb{Z}_q^m$, so $\mathbf{c}^T \beta$ is uniformly random in $\mathbb{Z}_q$ for a non-trivial $\mathbf{c}$ (we need primality of q for this).

If $q \gg Bl \log n$, we can distinguish the instance by simply checking whether $|\mathbf{c}^T \beta| \leq \alpha Bl \log n$.

Theorem 6.15. *Let n, m, q, r, l, k be positive integers, q be prime and $q \gg r(l \log n)^{3/2}$. Let $\alpha > 0$ be an absolute constant. There exists an algorithm that solves the decisional k -LWE $_{n,m,q,r}$ problem correctly except with probability at most $n^{-\Omega(l)}$ provided the following:*

- *Let $k/2 \leq l \leq n - k/2$. Then provided that the number of samples $m \geq (l \ln(qn)) \cdot \left(\frac{\alpha q n}{l}\right)^{k/2}$, the algorithm takes time $\tilde{O}\left(\sqrt{\binom{n}{l} q^l (n/l)^k}\right)$ time.*
- *Let $k \leq l \leq n$. Then provided that the number of samples $m \geq (l \ln(qn)) \cdot \left(\frac{\alpha n}{l}\right)^k$, the algorithm takes time $\tilde{O}\left(\sqrt{\binom{n}{l} q^l (n/l)^k}\right)$ time.*

Proof. (We prove the large q case; the small q case is similar.) Let $\mathbf{c}$ be the q -ary cover of size $O(l \log n)$ returned by the algorithm in Corollary 6.12.

Suppose the instance (A, β) is from the planted distribution with error vector $\mathbf{e} \in \mathbb{Z}_q^m$. The product $\mathbf{c}^T \mathbf{e} = \tilde{\mathbf{c}}^T \tilde{\mathbf{e}}$ where $\tilde{\mathbf{c}}, \tilde{\mathbf{e}} \in \mathbb{Z}_q^t$ are truncated versions of $\mathbf{c}, \mathbf{e}$ over the support of $\mathbf{c}$ of size $t \leq \|\mathbf{c}\|_1 \leq O(l \log n)$. It is well known that for the discrete Gaussian with width parameter r , a vector $\tilde{\mathbf{e}} \leftarrow D_{\mathbb{Z}_q, r}^t$ ($t \ll q$) satisfies $\|\tilde{\mathbf{e}}\| \leq B = r\sqrt{t}$ except with probability $\leq \exp(-\Omega(t)) = n^{-\Omega(l)}$. From this and above discussion, $|\mathbf{c}^T \beta| \leq O(r(l \log n)^{3/2})$.

If the instance is from the random distribution, we will have $|\mathbf{c}^T \beta| \leq O(r(l \log n)^{3/2})$ with probability $O(r(l \log n)^{3/2})/q$, which will be the main failure probability for our distinguisher. $\square$

Remark 6.9. *Note that the hardness of standard LWE from lattice problems was established by Regev [Reg05] for any $r > 2\sqrt{n}$ and $r < q$. On the other hand, Theorem 6.15 solves sparse LWE in a parameter regime which is only mildly more restrictive than this; eg. it solves it for $r = 3\sqrt{n}$ and $q \gg n^2 \log^{3/2} n$.*

¹¹ $\mathbf{c}, \mathbf{e}$ have entries in $\mathbb{Z}_q$, but for this purpose, we can assume them as integers in $[-\lfloor q/2 \rfloor, \lfloor q/2 \rfloor]$.

6.3 Solving sparse LPN

The noise distribution in the LPN problem (see Definitions 3.6, 4.6) does not have a property of bounded error with high probability. On the other hand, it has a very special property of uniform probability mass for non-zero noise. We will exploit this to design a different technique for distinguishing these instances; this will be analogous to distinguishing idea from [GHOS26].

6.3.1 Polynomial Anticoncentration

We will need a result on the anti-concentration of certain polynomials of random variables, similar to [GHOS26].

Let us define a ‘block-linear’ m -variate polynomial with respect to a partition of $[m]$.

Definition 6.10. (Definition 25 in [GHOS26]) Let m, T be integers. Let

$$P(x_1, \dots, x_m) = \sum_{S \subseteq [m]} a_S \prod_{i \in S} x_i$$

be a multilinear polynomial, where $a_S \in \mathbb{R}$. Let $\pi = (\pi_1, \dots, \pi_T)$ be a partition of $[m]$. For $S \subseteq [m]$, we say π shatters S if $|S \cap \pi_j| \leq 1$ for every $j \in [T]$. Define the block-linear restriction of P to π as

$$P_\pi(x_1, \dots, x_m) := \sum_{\substack{S \subseteq [m] \\ \pi \text{ shatters } S}} a_S \prod_{i \in S} x_i.$$

We say that P is π -block-linear if $P_\pi = P$.

The following result says that for a random equi-partition π of $[m]$, the block-linear restriction of a polynomial to π does not lose too many terms.

Lemma 6.16. (Lemma 26 in [GHOS26]) Let $m, T \in \mathbb{N}$ with $T \mid m$, and let $P(x_1, \dots, x_m)$ be a sum of μ multilinear monomials in the x_i ’s of degree at most T . Then for a uniformly random equipartition $\pi = (\pi_1, \dots, \pi_T)$ of $[m]$,

$$\Pr[\text{number of monomials in } P_\pi \geq 0.5 \cdot \mu \cdot \exp(-T)] \geq \exp(-2T).$$

Next, we build an anti-concentration result for block-linear polynomials. This will have an analogous proof as in Proposition 30 of [GHOS26], and we give the proof details in Appendix C.

Lemma 6.17. (similar to Proposition 30 in [GHOS26]) Let $\mathcal{D}$ be a distribution on $\mathbb{C}$ with probability density $f_{\mathcal{D}} \leq 1$, mean $\rho \in \mathbb{R}_{>0}$ and support contained in $\{x \in \mathbb{C} : |x| \leq 1\}$. Fix integers $m, T \geq 1$, a partition $\pi = (\pi_1, \dots, \pi_T)$ of $[m]$, and let $P(x_1, \dots, x_m)$ be any π -block-linear polynomial of degree $\leq T$. Define

$$\mu := \mathbb{E}_{X_i \stackrel{iid}{\sim} \mathcal{D}} [P(X_1, \dots, X_m)] = P(\rho, \rho, \dots, \rho).$$

Then for C_{anti} same as in Lemma C.2:

$$\Pr \left[|P(X_1, \dots, X_m)| < C_{anti}^{-10T} |\mu| \right] \leq 0.9^T.$$

Algorithm 3 DISTINGUISH-USING- q -ARY-COVERS($\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}, \mathcal{C}$)

Require: A k -LIN $_q$ instance $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$, a set of non-trivial q -ary covers $\mathcal{C}$ of $\mathcal{I}$, each of size at most $2T$, with $|\mathcal{C}| = \lceil 10N^\varepsilon \rceil$.

```
1: for  $s = 1, \dots, S := 10e^{4T}$  do
2:   Sample a random partition  $\pi$  of  $[m]$  into  $2T$  equally-sized parts;
3:   Let  $\mathcal{C}' \subseteq \mathcal{C}$  be the set of  $q$ -ary covers whose support is shattered by  $\pi$ ;
4:   if  $|\mathcal{C}'| \geq N^\varepsilon \cdot 0.1^T$  then
5:     break the loop;
6:   end if
7: end for
8: if the loop never broke then
9:   output FAIL;
10: end if
11: if  $|\mathcal{C}'| > N^\varepsilon \cdot 0.1^T$  then
12:   restrict it to  $N^\varepsilon \cdot 0.1^T$  arbitrary ones;
13: end if
14: Sample  $\zeta_1, \dots, \zeta_m \stackrel{iid}{\sim} \text{Unif}([0, 1])$ ;
15: Define  $\Psi_{\mathcal{C}'} := \sum_{\substack{\mathbf{c} \in \mathcal{C}' \\ c_i > 0}} \prod_{i \in [m]} \zeta_i b_i^{c_i}$ ;
16: if  $|\Psi_{\mathcal{C}'}| \geq N^{0.6\varepsilon}$  then
17:   output PLANTED
18: else
19:   output RANDOM
20: end if
```

6.3.2 Distinguishing using q -ary covers

Our technique will be to use the collection $\mathcal{C}$ of distinct q -ary covers obtained from Algorithm 2. We present algorithm DISTINGUISH-USING- q -ARY-COVERS (Algorithm 3) for this purpose.

Firstly, we observe that Algorithm 3 does not output FAIL with probability 0.99. Indeed, Lemma 6.16 guarantees that for a random equi-partition π , there are at least $5N^\varepsilon e^{-2T} \geq N^\varepsilon 0.1^T$ shattered terms of $\mathcal{C}$ with probability at least e^{-4T} , i.e. the loop breaks in any iteration with probability at least e^{-4T} . Thus the loop breaks in $10e^{4T}$ iterations with probability at least $1 - (1 - e^{-4T})^{10e^{4T}} \geq 0.99$. Hence, in the rest of the argument, we condition on the event that the algorithm does not output FAIL.

Now, notice that our output decision is based on the quantity

$$\Psi_{\mathcal{C}'} := \sum_{\substack{\mathbf{c} \in \mathcal{C}' \\ c_i > 0}} \prod_{i \in [m]} \zeta_i b_i^{c_i} = \sum_{\substack{\mathbf{c} \in \mathcal{C}' \\ c_i > 0}} \prod_{i \in [m]} \zeta_i \eta_i^{c_i} (z^{S_i})^{c_i} = \sum_{\substack{\mathbf{c} \in \mathcal{C}' \\ c_i > 0}} \prod_{i \in [m]} \zeta_i \eta_i^{c_i} \quad (26)$$

where ζ_i are i.i.d. random variables from $\text{Unif}([0, 1])$, $\eta_i \stackrel{iid}{\sim} \mathcal{LPN}_\rho$ in the planted case, $\eta_i \sim \text{Unif}(\Omega_q)$ in the random case, and the last equality follows from the q -ary cover property.

The distribution of η_i in both cases is uniform over points in $\Omega_q \setminus \{1\}$. This leads to a crucial observation: For any $s \in \mathbb{Z}_q \setminus \{0\}$, $\eta_i^s \equiv \eta_i$. We use this in (26) to get

$$\Psi_{\mathcal{C}'} = \sum_{\mathbf{c} \in \mathcal{C}'} \prod_{\substack{i \in [m] \\ \mathbf{c}_i > 0}} \zeta_i \eta_i \quad (27)$$

Now $\Psi_{\mathcal{C}'}$ is a multilinear polynomial in $\{\zeta_i \eta_i\}_{i \in [m]}$. This will help us to better analyze the expectation, and use the anti-concentration results.

$$\begin{aligned} \mathbb{E}[\Psi_{\mathcal{C}'}] &= \sum_{\mathbf{c} \in \mathcal{C}'} \mathbb{E} \left[\prod_{\substack{i \in [m] \\ \mathbf{c}_i > 0}} \zeta_i \eta_i \right] \\ &= \sum_{\mathbf{c} \in \mathcal{C}'} \prod_{\substack{i \in [m] \\ \mathbf{c}_i > 0}} \mathbb{E}[\zeta_i] \mathbb{E}[\eta_i] \\ &= \begin{cases} 0 & \text{if } \mathcal{I} \text{ is random} \\ \sum_{\mathbf{c} \in \mathcal{C}'} (\rho/2)^{\text{supp}(\mathbf{c})} \geq \sum_{\mathbf{c} \in \mathcal{C}'} (\rho/2)^{\|\mathbf{c}\|_1} & \text{if } \mathcal{I} \text{ is planted} \end{cases} \end{aligned}$$

Like the method of eigenvalue estimation, we rely on the hypothesis that $|\Psi_{\mathcal{C}'}|$ will be low in the random case and high in the planted case.

However, $\Psi_{\mathcal{C}'}$ is not a sum of independent random variables as the q -ary covers in $\mathcal{C}'$ might share edges. This does not allow us to use standard machinery for concentration of $\Psi_{\mathcal{C}'}$ around its mean.

For the random case, we are still able to prove an upper bound for $\Psi_{\mathcal{C}'}$ using the second moment method. For the planted case, we obtain a lower bound for $\Psi_{\mathcal{C}'}$ by proving anti-concentration around 0.

For the following two lemmas, we work with

$$P_{\mathcal{C}'}(x_1, \dots, x_m) := \sum_{\mathbf{c} \in \mathcal{C}'} \prod_{\substack{i \in [m] \\ \mathbf{c}_i > 0}} x_i$$

Following is the result for the random case:

Proposition 6.18. *Fix $\gamma > 0$ and assume that $T > \gamma \varepsilon \log N$, where N, ε, T are parameters in Algorithm 3. If $X_1, \dots, X_m$ are complex-valued independent mean-zero random variables such that $|X_i| \leq 1$, then:*

$$\Pr \left[|P_{\mathcal{C}'}(X_1, \dots, X_m)| < N^{0.6\varepsilon} \right] \geq 1 - N^{-\Omega(\varepsilon)}.$$

Proof. We show this using the second moment method. Firstly, $\mathbb{E}[P_{\mathcal{C}'}(X_1, \dots, X_m)] = 0$ by the fact that $P_{\mathcal{C}'}$ is multilinear and X_i 's are independent. Further, Corollary 6.9 guarantees that (except with probability $\exp(-N^\varepsilon)$) for any distinct $\mathbf{c}, \mathbf{c}' \in \mathcal{C}'$, $\mathbf{c}$ contains some singleton constraint which is not in $\mathbf{c}'$. This implies that the summands in $P_{\mathcal{C}'}(X_1, \dots, X_m)$ are pairwise independent. Thus

$$\mathbb{E} \left[|P_{\mathcal{C}'}(X_1, \dots, X_m)|^2 \right] = \mathbb{E} \left[\sum_{\mathbf{c} \in \mathcal{C}'} \prod_{\substack{i \in [m] \\ \mathbf{c}_i > 0}} |X_i|^2 \right] \leq |\mathcal{C}'| = N^{\varepsilon 0.1 T}$$

where the second last inequality is by the bound on X_i 's and the last inequality follows from line 11 in Algorithm 3. We complete the argument by applying Chebyshev's inequality:

$$\begin{aligned} \Pr [|P_{C'}(X_1, \dots, X_m)| \geq N^{0.6\varepsilon}] &\leq \frac{\text{Var}(P_{C'}(X_1, \dots, X_m))}{N^{1.2\varepsilon}} \leq \frac{N^{\varepsilon} 0.1^T}{N^{1.2\varepsilon}} \\ &= N^{-0.2\varepsilon} 0.1^T \leq N^{-(0.2+2\gamma)} \end{aligned}$$

□

Now we show a lower bound for the planted case:

Proposition 6.19. *Let $\mathcal{D}$ be a distribution on $\mathbb{C}$ with probability density $f_{\mathcal{D}} \leq 1$, mean $\rho \in \mathbb{R}_{>0}$ and support contained in $\{x \in \mathbb{C} : |x| \leq 1\}$. Fix $\gamma > 0$, and assume $\gamma\varepsilon \log N < T < \frac{0.4\varepsilon \log N}{\log(10(C_{\text{anti}})^{20}\rho^{-4})}$, where $2T$ is the bound on the cover sizes in $\mathcal{C}'$. Then:*

$$\Pr_{X_1, \dots, X_m \stackrel{iid}{\sim} \mathcal{D}} [|P_{C'}(X_1, \dots, X_m)| \geq N^{0.6\varepsilon}] \geq 1 - N^{-\Omega(\varepsilon)}.$$

Proof. Note that $\mathcal{C}'$ consists of $N^{\varepsilon} 0.1^T$ q -ary covers of size $2T$. This give us $\mathbb{E}[P_{C'}(X_1, \dots, X_m)] = P_{C'}(\rho/2, \dots, \rho/2) = |\mathcal{C}'| (\rho/2)^{2T} = N^{\varepsilon} 0.1^T (\rho/2)^{2T}$.

By Lemma 6.17, we have

$$\Pr [|P_{C'}(X_1, \dots, X_m)| < C_{\text{anti}}^{-20T} N^{\varepsilon} 0.1^T (\rho/2)^{2T}] \leq 0.9^{2T} \leq N^{-0.1\gamma\varepsilon}$$

Finally the result follows by using the upper bound assumed on T on the upper bound of $|P_{C'}(X_1, \dots, X_m)|$ obtained above.

□

With these results, we are now ready to prove the following:

Theorem 6.20. *Let n, m, q be positive integers, q be prime, $0 \leq \mu \leq \frac{q-1}{q} - \frac{1}{n^d}$ for some constant $d > 0$. Let $\alpha > 0$ be an absolute constant. Also, let $\varepsilon > 0$ be a tolerance parameter. There exists an algorithm that solves the decisional k -LPN $_{n,m,q,\mu}$ problem such that:*

- *Let $l \in \mathbb{N}$ be a parameter such that $k/2 \leq l \leq n - k/2$. Then provided that the number of samples $m \geq n^{2d}(l \ln(qn)) \cdot (\frac{\alpha n}{l})^{k/2}$, the algorithm takes time $\tilde{O}\left(\left(\binom{n}{l} q^l\right)^{1/2+\varepsilon} \left(\frac{n}{l}\right)^k\right)$ time and is correct except with probability at most $n^{-\Omega(\varepsilon l)}$.*
- *Let $l \in \mathbb{N}$ be a parameter such that $k \leq l \leq n$. Then provided that the number of samples $m \geq n^{2d}(l \ln(qn)) \cdot (\frac{\alpha n}{l})^k$, the algorithm takes time $\tilde{O}\left(\left(\binom{n}{l} q^l\right)^{1/2+\varepsilon} \left(\frac{n}{l}\right)^k\right)$ and is correct except with probability at most $n^{-\Omega(\varepsilon l)}$.*

Proof. (We prove the large q case; the small q case is similar.) By Propositions 6.18, 6.19 we get that for some arbitrary constant $\gamma > 0$, $\gamma\varepsilon \log N < T < \frac{0.4\varepsilon \log N}{\log(10(C_{\text{anti}})^{20}\rho^{-4})}$, Algorithm 3, given as input a set $\mathcal{C}$ of nonempty q -ary covers of size $\leq 2T$ in $\mathcal{I}$, with $|\mathcal{C}| \geq 10N^{\varepsilon}$, outputs a guess **RANDOM** or **PLANTED**, correct with probability at least $1 - N^{-\Omega(\varepsilon)}$. Using this after applying Algorithm 2 on the Kikuchi Graph $\mathcal{K}_l(\mathcal{I})$ (whose correctness is given by Corollary 6.14) establishes the correctness of the algorithm.

Corollary 6.14 also establishes the time complexity of Algorithm 2 which matches the claimed time complexity. We now show that the time of Algorithm 3 is dominated by this time complexity.

The `for` loop in steps 1-7 makes $e^{4T} = N^{O(\varepsilon)}$ iterations. Within each iteration, step 2 takes $\tilde{O}(m)$ time and steps 3-6 are doable in $O(|\mathcal{C}|T) = \tilde{O}(N^\varepsilon)$ time, with a total time of $N^{O(\varepsilon)}\tilde{O}(m)$. The remaining steps are doable in $\tilde{O}(|\mathcal{C}'|T) = \tilde{O}(N^{0.6\varepsilon})$ time. Overall, this algorithm takes $N^{O(\varepsilon)}\tilde{O}(m) = \tilde{O}(m \binom{n}{l} q^l)^{O(\varepsilon)}$ time. $\square$

7 Discussion on Sample-Time Tradeoffs

This work aims to explore algorithms against sparse LWE/LPN for higher modulus q which yield good sample complexity-time complexity tradeoffs. In this section, we discuss our attacks as well as those proposed in [JLS24].

7.1 Comparing the spectral method and the q -ary cover method

In Sections 5, 6 we presented two algorithms and parameterized the sample-time tradeoffs for both of them. The q -ary cover method (see Theorems 6.15, 6.20) has the following perks compared to the spectral method (see Theorems 5.20, 5.21):

- For $l \gg k$ it enjoys a nearly quadratic speedup for the same sample complexity compared to the spectral method.
- It works when the k -sparse left-hand sides have *arbitrary* distribution (not necessarily from $D_{\text{coeff},n,k,q}$). Unlike this, the spectral method relies on the fact that the left-hand sides are uniformly random (see Proposition 5.5).

On the other hand, the spectral method too enjoys several perks:

- The spectral method does not assume any condition on q (either prime or lower bounded). Unlike this, we needed q to be prime for our analysis for q -ary method. Apart from that, we needed $q \gg r(l \log n)^{3/2}$ (r was the width of the discrete Gaussian) for the sparse LWE attack via q -ary cover method.
- The spectral method is oblivious to the error model in the sense that it works for any noise distribution on the planted instance as long as the noise is sufficiently concentrated on zero and is symmetric about 0. Again, we are only able to analyze the q -ary method for bounded-error distributions and the LPN distribution.
- The spectral method has a quantum analog with a speedup ranging from quadratic to quartic, depending on the parameter l .

7.2 Attacks by [JLS24]

Jain, Lin, Saha [JLS24] proposed two attack strategies on sparse LWE and demonstrated that these attacks take exponential time with sample complexity $m = \text{poly}(n)$ and sparsity $k = \Omega(\log n)$. Let us discuss them and see how they compare with our attacks. In this subsection, we denote a sparse LWE instance by $(A, \beta) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$ for a prime modulus q , where the rows of A, β correspond to the samples.

Dense Minor Lower-Bound Model: In dense minor attack, the idea was to find a set of rl rows in A (for some $r > 1$) all of which have non-zero entries only at a fixed set of l columns. If

we can find such a set of rows $A' \in \mathbb{Z}_q^{l \times n}$ (and let β' be the vector corresponding to the RHS of these rows), without loss, we can remove the zero columns and view A' as having l columns, and the instance (A', β') can be viewed as a k -sparse LWE instance of dimension l . We can then apply any known LWE-solving algorithm, such as the Arora-Ge algorithm [AG11], BKW algorithm or variants [BKW03; Lyu05; ACFFP15], or a lattice-based algorithm to solve this instance in time 2^{cl} for some constant $c > 0$.

We perform a calculation similar to the proof of Theorem 7.3 of [JLS24] (details in Appendix D) to get a lower bound on the sample complexity for feasibility of this attack.

$$m \geq \frac{l^2 (n/l)^k}{e^k n l^{1/l}} \approx l (n/el)^{k-1} \quad (28)$$

Although this matches our lower bound from the spectral and q -ary cover attacks, we emphasize that this is a *necessary* lower bound (i.e. an actual lower bound could be much more), whereas ours was a *sufficient* lower bound.

Sparse vectors in the Kernel: In this approach, one had to find a l -sparse vector $\mathbf{x} \in \mathbb{Z}_q^m$ such that $\mathbf{x}^T A = 0$. So, if (A, β) is a planted LWE instance, i.e. $\beta = A\mathbf{s} + \mathbf{e}$, we have $\mathbf{x}^T \beta = \mathbf{x}^T \mathbf{e}$. Further, if we assume (with high probability) $\mathbf{e} \in [-B, B]^n$ for some bound B , we can search for an $\mathbf{e}$ by brute force that satisfies $\mathbf{x}^T \beta = \mathbf{x}^T \mathbf{e}$. On the other hand, $\mathbf{x}^T \beta$ will be uniformly random when β is uniformly random, and so with high probability will not be equal to $\mathbf{x}^T \mathbf{e}$ for any bounded $\mathbf{e}$ when q is large enough. This takes $B^{O(l)}$ time, as we only need to check for coordinates of $\mathbf{e}$ corresponding to the non-zero coordinates in $\mathbf{x}$.

We remark that this is almost the same attack as the q -ary cover method for sparse LWE! Indeed, we had seen in Subsection 6.2 that $\mathbf{c}$ is a q -ary cover iff it satisfies $\mathbf{c}^T A = 0$. The only difference is that we are imposing a stronger condition on $\mathbf{c}$ than l -sparse; $\mathbf{c}$ must satisfy $\|\mathbf{c}\|_1 \leq O(l \log n)$. We are able to find such a $\mathbf{c}$ using the Kikuchi Graph (we emphasize that [JLS24] had only analyzed the feasibility of this attack *assuming* one has such a vector $\mathbf{c}$). Moreover, the stronger condition on $\mathbf{c}$ also slightly simplifies our strategy afterward; we need only check $|\mathbf{c}^T \beta|$ instead of searching for an error vector via brute-force. We stated the sample-time tradeoff for this in Theorem 6.15.

7.3 Conjectured Hardness of sparse LWE

As an interesting case, let us analyze the sample complexity required if we allow a time of $2^{\delta n}$ for an arbitrarily small $\delta > 0$. We have $m \geq \left(\frac{\Omega(n)}{l}\right)^k$ samples required for a time of $\binom{n}{l} q^l$ (say for the spectral method). We must have

$$\begin{aligned} \binom{n}{l} q^l \approx 2^{\delta n} &\Rightarrow l \approx \frac{\delta n}{\log q + \log n} \\ \Rightarrow m = O\left((\ln n) (\log q + \log n)^k\right) &\text{ samples are sufficient} \end{aligned}$$

In particular, if $k = O(\log n / \log \log n)$ then polynomially many samples suffice. If $k = O(\log n)$, then we require $(\log n)^{O(\log n)}$ samples which is slightly more than polynomial.

Conjecture 7.1 in [JLS24] suggests that k -LWE $_{n,m,q,r}$ is as hard as standard LWE over $\Theta(n)$ dimension, $\text{poly}(n)$ samples and the same modulus and width parameters, when the sparsity $k = \Omega(\log n)$ and $m = \text{poly}(n)$. Assuming hardness of standard LWE, the algorithm from the Kikuchi method nearly closes the gap of this conjecture, either when the sparsity is slightly less, or the polynomial samples requirement is slightly relaxed.

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A Relation of ρ with the width r of discrete Gaussian

Let us see how the expectation of ρ in the $\mathcal{LWE}_\rho$ distribution (Definition 4.7) depends on the width parameter r of the discrete Gaussian.

Definition A.1. For $\rho \in [0, 1]$, let $\mathcal{LWE}_\rho$ be the distribution on Ω_q such that for $\eta \sim \mathcal{LWE}_\rho$, $\mathbb{E}[\eta] = \rho$ and $\eta = \omega_q^e$ where $e \sim D_{\mathbb{Z}_q, r}$. This means that $r \in [0, \infty]$ is the (unique) solution to the equation

$$\rho = \mathbb{E}[\eta_i] = \frac{\sum_{j \in \mathbb{Z}_q} \omega_q^j \sum_{x \in q\mathbb{Z}+j} e^{-\pi x^2/r^2}}{\sum_{x \in \mathbb{Z}} e^{-\pi x^2/r^2}} \quad (29)$$

$$= \frac{\sum_{x \in \mathbb{Z}} \omega_q^x e^{-\pi x^2/r^2}}{\sum_{x \in \mathbb{Z}} e^{-\pi x^2/r^2}} = \frac{\sum_{x \in \mathbb{Z}} \cos\left(\frac{2\pi x}{q}\right) e^{-\pi x^2/r^2}}{\sum_{x \in \mathbb{Z}} e^{-\pi x^2/r^2}} \quad (30)$$

Remark A.2. Let us obtain a lower bound for (30). Recall the Poisson Summation Formula for $f : \mathbb{R} \rightarrow \mathbb{R}$ where $\hat{f}$ denotes its Fourier transform:

$$\sum_{x \in \mathbb{Z}} f(x) = \sum_{y \in \mathbb{Z}} \hat{f}(y)$$

For $g(x) = \cos\left(\frac{2\pi x}{q}\right) e^{-\pi x^2/r^2}$, $\hat{g}(y) = \frac{r}{2} \left(e^{-\pi r^2 \left(y - \frac{1}{q}\right)^2} + e^{-\pi r^2 \left(y + \frac{1}{q}\right)^2} \right)$ and for $h(x) = e^{-\pi x^2/r^2}$, $\hat{h}(y) = r e^{-\pi r^2 y^2}$. Using these in (30), we obtain

$$\rho = \frac{\sum_{y \in \mathbb{Z}} \hat{g}(y)}{\sum_{y \in \mathbb{Z}} \hat{h}(y)} = \frac{e^{-\pi r^2/q^2} \sum_{y \in \mathbb{Z}} \cosh\left(\frac{2\pi r^2 y}{q}\right) e^{-\pi r^2 y^2}}{\sum_{y \in \mathbb{Z}} e^{-\pi r^2 y^2}} \geq e^{-\pi r^2/q^2}$$

In our results, we see that the sample complexity required for both distinguishing methods using Kikuchi graphs depends inverse polynomially on ρ . Since the width r for the LWE distribution is chosen to be less than q , this dependency will only contribute a constant term.

B A quantum speedup

Some Notation. $\mathbf{x}^\dagger$ denotes the conjugate transpose of a vector $\mathbf{x}$. $|\mathbf{x}\rangle$ denotes the unit vector in the direction of $\mathbf{x}$ and $\langle \mathbf{x} | := |\mathbf{x}\rangle^\dagger$. For a matrix A , $\|A\|_{\max} := \max_{i,j} |A_{ij}|$.

In essence, to solve decisional k -LIN $_q$, the Kikuchi method performs estimation of the spectral norm of a sparse Hermitian matrix in a space of dimension $N = \binom{n}{l} q^l$. For any known classical method, such as the power iteration, eigenvalue estimation requires matrix multiplications, so it is unlikely for the time complexity to go below $\tilde{\Omega}(N)$. However, with quantum algorithmic techniques, we can do better.

A Hamiltonian H on t qubits is a Hermitian matrix of dimension 2^t . H is called s -sparse if each row and each column of H has at most s non zero entries. Throughout the discussion, we shall assume the normalization constraint that $\|H\|_{\max} \leq 1$. Approximating the spectral norm of a sparse Hamiltonian is well studied in quantum computing. Indeed, for a local Hamiltonian (which is a special case of sparse hamiltonian), this problem is QMA complete. However, if we are

provided with a *guiding* state $|\gamma\rangle$, i.e. a state that has good overlap with a unit eigenvector $|v\rangle$ corresponding to the maximum absolute eigenvalue, i.e. $(\langle\gamma|v\rangle)^2$ is large, the problem can be easier. If we are provided with a guiding state having $1/\text{poly}(t)$ overlap with the leading eigenvector, then Gharibian and Le Gall [GLG22] show that the problem is BQP-complete.

Schmidhuber et al. [SOKB25] provide a quantum algorithm to solve the Guided Sparse Hamiltonian problem by combining standard quantum algorithmic techniques of Hamiltonian simulation, phase estimation and amplitude estimation. We define the problem and state their result below. For details, the reader may refer to their paper.

B.1 Guided Sparse Hamiltonian Problem

The input Hamiltonian H is assumed to be given in the form of quantum oracle access by providing oracles O_H and O_F . O_H is called the *adjacency matrix* oracle, which on input (i, j) (for $i, j \in [2^t]$) gives the entry H_{ij} of H upto rounding. O_F is the *adjacency list* oracle, which on input (i, r) (for $i \in [2^t], r \in [s]$) either returns the index of the r^{th} non-zero entry in the i^{th} row, or indicates that there are less than r non-zero entries in i^{th} row.

The advantage of having oracle access to H in this manner is that we do not have to store the Hamiltonian H that may possibly be of large dimension; rather we can access it “on the fly” when required.

Definition B.1. (Definition 62 in [SOKB25]) *In the GSHS problem, we are given the following as input :*

1. An s -sparse hamiltonian H on t qubits with $\|H\|_{\max} \leq 1$ as specified with the oracles O_H and O_F .
2. A quantum circuit using G gates which performs the mapping : $|0^t\rangle |0^A\rangle \mapsto |\Psi\rangle |0^A\rangle$.
3. Parameters $\lambda \in \mathbb{R}, \alpha \in (0, 1), \gamma \in (0, 1]$.

To solve it, one must output :

- YES, if $\|\Pi_{\geq \lambda}(H) |\Psi\rangle\| \geq \gamma$ (which also implies $\|H\| \geq \lambda$)
- NO, if $\|H\| \leq (1 - \alpha)\lambda$

Theorem B.1. (Theorem 63 in [SOKB25]) *The Guided Sparse Hamiltonian problem can be solved with high probability by a quantum algorithm that uses:*

- $(Q = \tilde{O}(s/\gamma\alpha\lambda))$ queries to the oracles
- $\tilde{O}(G/\gamma + \text{polylog}(Q)/\gamma + Qt)$ gates
- $(A + O(t) + \tilde{O}(\log Q))$ qubits

For a $k\text{-LIN}_q$ instance $\mathcal{I}$ of dimension n and m samples, the associated Hermitian matrix $\mathcal{K}_l(\mathcal{I})$ has the following parameters associated with it:

1. $2^t = N = \binom{n}{l}(q^l)$, so $t = O(l \log n)$
2. Its sparsity $s = \Delta = \frac{m\theta^{(n,l,k)}}{\binom{n}{l}(q^l)} = O(\log N) = O(l \log n)$ for the separation in spectral norm for random vs planted case (from Propositions 5.9, 5.15)

3. $\lambda > (1 + O(1))\rho\Delta$, say $\lambda = 3/2\rho\Delta$ (lower bound of spectral norm for planted case)
4. $(1 - \alpha)\rho\Delta = (1 - O(1))\rho\Delta$, say $\alpha = 1/2$ (upper bound of spectral norm for random case)

Additionally, the oracles O_H and O_F can be efficiently computed using Construction 4.9.

From point 3. of Definition B.1, γ^2 is the parameter which corresponds to the overlap of the guiding state $|\Psi\rangle$. If γ is less than inverse polynomial, then from points 1 and 2 of Theorem B.1, $1/\gamma$ is the dominating factor for the time (i.e. gate) complexity.

Note that if $|\Psi\rangle$ is a random unit vector, then its overlap (with high probability) to the cutoff eigenspace (corr. to $\Pi_{\geq \lambda}$ that in the worst case is of small dimension) will be inversely proportional to the dimension with high probability, i.e.

$$\gamma^2 = \Omega\left(\frac{1}{\binom{n}{l} q^l}\right)$$

in our setting. This already yields a quantum algorithm with complexity $\frac{\text{poly}(n)}{\gamma} = \text{poly}(n) \cdot \sqrt{\binom{n}{l} q^l}$, a (nearly) quadratic speedup over the classical algorithm.

The other key observation of [SOKB25] is the ability to prepare a better guiding state, one which has overlap of $\Omega\left(\frac{1}{\sqrt{\binom{n}{l} q^l}}\right)$ with the cutoff-eigenspace.

B.2 Intuition for Guiding State

Again, we give a brief intuition behind the better guiding state; the details of the proofs will be analogous to Appendix C of [SOKB25].

Suppose for an instance $\mathcal{I} = \{(S_i, b_i)\}_{i \in [m]}$ we construct l/k copies of the following state:

$$|\gamma(\mathcal{I})\rangle := \frac{1}{\sqrt{m}} \sum_i b_i |S_i\rangle \in \mathbb{C}^{n^k q^k}$$

This state correlates well (better than random) with the planted assignment $|\mathbf{z}\rangle = \frac{1}{\sqrt{n}} \sum_i z_i |i\rangle$:

$$|\langle \mathbf{z}^{\otimes \ell} | \gamma(\mathcal{I})^{\otimes \ell/k} \rangle|^2 = \left(\langle \mathbf{z}^{\otimes k} | \gamma(\mathcal{I}) \rangle \right)^{2\ell/k} \approx \left(\frac{\rho \cdot m}{n^k q^k} \right)^{\ell/k}$$

If $m \approx \Theta((qn)^{k/2})$ then the overlap $\sim (qn)^{-l/2}$, which is a quadratic improvement from that of a random guiding state. The actual guiding state is approximately a symmetrized and a normalized version of $|\gamma(\mathcal{I})\rangle$. Also, the other technical challenge is to show a high overlap with the cutoff eigenspace of $\mathcal{K}_l(\mathcal{I})$ rather than with $\mathbf{z}$. These issues are handled in detail in Appendix C of [SOKB25].

C Polynomial Anticoncentration

To prove an anti-concentration result for block-linear polynomials, we first develop anti-concentration results for linear polynomials.

Lemma C.1. (Lemma 27 in [GHOS26]) Let $X_1, \dots, X_n$ be independent random variables, where each X_i is mean-zero, supported on $[-2, 2]$, and with p.d.f. bounded by 1. Let $L = c_1X_1 + \dots + c_nX_n$, where $c_1, \dots, c_n \in \mathbb{R}$. Then for all $0 < \varepsilon < 1/2$, we have

$$\Pr[|L - 1| \leq \varepsilon] \leq C'_{anti} \cdot \varepsilon \sqrt{\log(1/\varepsilon)},$$

where $1.2 < C'_{anti} < \infty$ is a universal constant.

We can easily extend this to the case when the random variables X_i are complex-valued.

Lemma C.2. Let $X_1, \dots, X_n$ be independent random variables, where each X_i is complex valued, is bounded as $|X_i| \leq 2$, has mean 0 and has p.d.f. bounded by 1. Let $L = c_1X_1 + \dots + c_nX_n$, where $c_1, \dots, c_n \in \mathbb{R}$. Then for all $0 < \varepsilon < 1/2$, we have

$$\Pr[|L - 1| \leq \varepsilon] \leq C_{anti} \cdot \varepsilon \sqrt{\log(1/\varepsilon)},$$

where $C_{anti} < \infty$ is a universal constant.

Proof. We can view a complex-valued random variable as a random variable in $\mathbb{R}^2$; let $X_i = (X_{i,re}, X_{i,im})$ for $X_{i,re}, X_{i,im} \in \mathbb{R}$. Then, $\mathbb{E}[X_{i,re}] = 0$ and $\{X_{i,re}\}_{i \in [m]}$ are independent. Further, the marginal p.d.f. of $X_{i,re}$ is bounded as

$$\begin{aligned} f_{X_{i,re}}(x) &= \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} f_{X_i}(x, y) dy \quad (f_{X_i} \text{ denotes the p.d.f. of } X_i) \\ \Rightarrow f_{X_{i,re}}(x) &\leq 2\sqrt{4-x^2} \leq 4 \quad (\text{using } f_{X_i} \leq 1) \end{aligned}$$

Now, $L = L_{re} + \iota L_{im}$ where $L_{re} = c_1X_{1,re} + \dots + c_nX_{n,re}$ and we have

$$\Pr[|L - 1| \leq \varepsilon] \leq \Pr[|L_{re} - 1| \leq \varepsilon]$$

and the result follows by applying Lemma C.1 (the same proof will work with p.d.f. bound 4 instead of 1). $\square$

As a next step, we prove an anti-concentration result for affine polynomials.

Corollary C.3. (similar to Corollary 29 in [GHOS26]) Let $X_1, \dots, X_m$ be independent random variables with mean $\rho \in \mathbb{R}_{>0}$, supported on $\{x \in \mathcal{C} : |x| \leq 1\}$, and with pdf bounded by 1. Let L be an affine polynomial on n variables with real coefficients. Then for all $0 < \varepsilon < 1/2$, we have

$$\Pr[|L(X_1, \dots, X_m)| \leq \varepsilon L(\rho, \dots, \rho)] \leq C_{anti} \cdot \varepsilon \sqrt{\log(1/\varepsilon)}$$

(where C_{anti} is the constant from Lemma 27).

Proof. (same proof in [GHOS26]) Note that by affineness,

$$\begin{aligned} L(x_1, \dots, x_m) &= L(x_1 - \rho, \dots, x_m - \rho) + L(\rho, \dots, \rho) - L(0, \dots, 0) \\ L(x_1 - \rho, \dots, x_m - \rho) - L(0, \dots, 0) &\text{ is a homogeneous linear polynomial} \end{aligned}$$

Hence we can apply Lemma C.2 to obtain

$$\Pr[|L(X_1 - \rho, \dots, X_m - \rho) - L(0, \dots, 0)| \leq \varepsilon L(\rho, \dots, \rho)] \leq C_{anti} \cdot \varepsilon \sqrt{\log(1/\varepsilon)}.$$

This event is equivalent to the desired event. $\square$

With Lemma C.2 and Corollary C.3, the following lemma can also be proved in exactly the same way as Proposition 30 in [GHOS26].

Lemma C.4. (similar to Proposition 30 in [GHOS26]) Let $\mathcal{D}$ be a distribution on $\mathbb{C}$ with probability density $f_{\mathcal{D}} \leq 1$, mean $\rho \in \mathbb{R}_{>0}$ and support contained in $\{x \in \mathbb{C} : |x| \leq 1\}$. Fix integers $m, T \geq 1$, a partition $\pi = (\pi_1, \dots, \pi_T)$ of $[m]$, and let $P(x_1, \dots, x_m)$ be any π -block-linear polynomial of degree $\leq T$. Define

$$\mu := \mathbb{E}_{X_i \stackrel{\text{iid}}{\sim} \mathcal{D}} [P(X_1, \dots, X_m)] = P(\rho, \rho, \dots, \rho).$$

Then for C_{anti} same as in Lemma C.2:

$$\Pr \left[|P(X_1, \dots, X_m)| < C_{\text{anti}}^{-10T} |\mu| \right] \leq 0.9^T.$$

Proof. (Sketch) Without loss of generality, we assume that for $t \in [T]$, $\pi_t = \{(t-1)\frac{m}{T} + 1, \dots, t\frac{m}{T}\}$. For $t \in [T] \cup \{0\}$ we define the random variable

$$R_t = P(X_1, \dots, X_{tm/T}, \rho, \dots, \rho)$$

By the fact that P is π -block linear, $R_{t+1}|X_1, \dots, X_{tm/T}$ is an affine polynomial in variables $\{X_{tm/T+1}, \dots, X_{(t+1)m/T}\}$, and $\mathbb{E}[R_{t+1}|X_1, \dots, X_{tm/T}] = R_t$. We can then apply Corollary C.3 to this polynomial to get:

$$\Pr \left[|R_{t+1}| \leq \varepsilon |R_t| \mid X_1, \dots, X_{tm/T} \right] \leq C_{\text{anti}} \cdot \varepsilon \sqrt{\log(1/\varepsilon)} \quad (31)$$

We can use (31) to get the following expectation (start with $E[Y] = \int_0^\infty \Pr[Y > y] dy$ for this)

$$\mathbb{E} \left[\left| \frac{R_t}{R_{t+1}} \right|^{0.1} \mid X_1, \dots, X_{tm/T} \right] \leq 1 + 0.04 C_{\text{anti}} \quad (32)$$

Next, we will use (32) telescopically to get

$$\mathbb{E} \left[\left| \frac{R_0}{R_T} \right|^{0.1} \right] \leq (1 + 0.04 C_{\text{anti}})^T \quad (33)$$

Note that $R_0 = \mu$ and $R_T = P(X_1, \dots, X_m)$. We get the desired result by applying Markov's inequality to (33) (and choosing a sufficiently large value of C_{anti}). $\square$

D Sample Complexity Calculations for Dense Minor

Let us calculate the sample complexity m required for the existence of a dense minor to exist with high probability. Our analysis closely resembles that in the proof of Theorem 7.3 of [JLS24]. Let us analyze the probability of existence of $rt = t + 1$ rows supported on a set of t columns for $k \leq t \leq l$. (To apply an LWE-style attack, we'll actually need $r > 1 + 1/t$, but the lower bound from this necessary condition suffices for our purpose). We can choose a set of t columns in $\binom{n}{t}$ ways, and choose $t + 1$ rows in $\binom{m}{t+1}$ ways. The probability of a set of rows having non-zero entries in those t

columns is given by $\left(\frac{\binom{t}{k}}{\binom{n}{k}}\right)^t$. Thus by a union bound, the required probability $p_{\text{dense_minor}}$ is upper bounded by:

$$p_{\text{dense_minor}} \leq \sum_{t=k}^l \binom{n}{t} \binom{m}{t+1} \left(\frac{\binom{t}{k}}{\binom{n}{k}}\right)^{t+1} \quad (34)$$

$$\leq l \binom{n}{l} \binom{m}{l+1} \left(\frac{\binom{l}{k}}{\binom{n}{k}}\right)^{l+1} \quad (\text{assuming } l \ll n, m) \quad (35)$$

$$\leq l e^{2l+1+k(l+1)} \left(\frac{n}{l}\right)^l \left(\frac{m}{l+1}\right)^{l+1} \left(\frac{l}{n}\right)^{k(l+1)} \quad (36)$$

$$\approx \exp(\ln l + kl + l \ln n + l \ln m - kl \ln(n/l) - 2l \ln l) \quad (37)$$

To be able to apply the dense minor attack, we must have existence of a dense minor (i.e. $l+1$ rows supported on a fixed set of l columns) with non-negligible probability. Thus (37) must be non-negligible, which gives us the bound:

$$m \geq \frac{l^2 (n/l)^k}{e^k n l^{1/l}} \approx l (n/el)^{k-1} \quad (38)$$